

THE RANKER'S MASTERBOOK

TRIG.

The Circular Arsenal

Complete Solutions

106

PROBLEMS

10

CHAPTERS

3–5

DIFFICULTY

212

WORKED SOLUTIONS

sin

106 original trigonometry problems where the identity is only half the battle – for the very top of JEE Advanced and the Mathematical Olympiad. Every problem solved more than one way.

Contents

Ten chapters, 106 problems.

I	Conditional Angle Identities · $A+B+C$	Nº1–10
II	Product-to-Sum Telescopes · Π	Nº11–20
III	Unit Roots & Finite Trig Sums · ζ_n	Nº21–30
IV	Chebyshev & Multiple Angles · T_n	Nº31–40
V	Trigonometric Substitution Engines · $x=\tan\theta$	Nº41–51
VI	Concavity, Tangents & Bounds · $\leq$	Nº52–61
VII	Phase, Range & Extremum Machines · $\mathbf{R}$	Nº62–71
VIII	Periodicity & Root-Counting · $=$	Nº72–86
IX	Triangle Geometry Through Trig · $\triangle$	Nº87–96
X	Grand Hybrids · $\oplus$	Nº97–106

CHAPTER I

Conditional Angle Identities $A+B+C$

Constrained angles turn symmetric sums into identities.

№1 The Tangent Product Lock

●●●●● L3

EVALUATE

If $A + B + C = \pi$, evaluate $\frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C}$.

angle-sum tangent symmetric

ANSWER

1

THE TRAP

The seductive wrong answer is 0, from reading $\tan(A + B + C) = 0$ as a zero numerator. The tangent addition formula has the denominator too; the zero is exactly the equality of numerator and product.

Method I: The chapter instrument

1. Apply $\tan(A + B + C) = 0$ and expand the tangent addition formula; since the denominator is finite in the stated expression, the numerator equals the triple product.
2. Substituting the given restrictions and simplifying gives 1.

Method II: Independent check

1. Start from $\sin(A + B + C) = 0$, divide by $\cos A \cos B \cos C$, and the same tangent relation appears without memorising the formula.
2. This route reaches the same value and rules out the trap, so the answer is 1.

INSIGHT

For triangle angles, tangent identities usually come from dividing a sine identity by a product of cosines.

№2 Half-Angle Gate

●●●●● L3

EVALUATE

If $A + B + C = \pi$ and $x = \tan \frac{A}{2}$, $y = \tan \frac{B}{2}$, $z = \tan \frac{C}{2}$, find $xy + yz + zx$.

half-angle constraint tangent

ANSWER

1

THE TRAP

The tempting answer is xyz , copied from the full-angle tangent formula. But the half-angles sum to $\pi/2$, where the tangent denominator vanishes instead.

Method I: The chapter instrument

1. Use $\frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$; for three angles with sum $\pi/2$, $1 - xy - yz - zx = 0$.

2. Substituting the given restrictions and simplifying gives 1.

Method II: Independent check

1. Write $\tan(u + v + w)$ and let $u + v + w \rightarrow \pi/2$; the denominator $1 - xy - yz - zx$ must be zero.
2. This route reaches the same value and rules out the trap, so the answer is 1.

INSIGHT

The half-angle substitution turns triangle constraints into the algebraic surface $xy + yz + zx = 1$.

№3 The Cotangent Mirror

●●●●● L3

EVALUATE

If $A + B + C = \pi$, evaluate $\cot A \cot B + \cot B \cot C + \cot C \cot A$.

cotangent

triangle

symmetric

ANSWER

1

THE TRAP

A common slip is 0, caused by replacing cotangents with tangents termwise. Reciprocals change the tangent product identity into a pairwise cotangent identity.

Method I: The chapter instrument

1. Divide $\sin(A + B + C) = 0$ by $\sin A \sin B \sin C$ to get the sum of pairwise cotangent products.
2. Substituting the given restrictions and simplifying gives 1.

Method II: Independent check

1. Invert the tangent relation $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ after dividing by $\tan A \tan B \tan C$.
2. This route reaches the same value and rules out the trap, so the answer is 1.

INSIGHT

Cotangent is often the cleaner language when sine products are present.

№4 Cosine Squares With a Hidden Constant

●●●●● L4

EVALUATE

If $A + B + C = \pi$, simplify $\cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C$.

cosine

identity

triangle

ANSWER

1

THE TRAP

The trap is 3, from bounding each cosine square by 1 and mistaking an inequality for an identity. The cubic term is exactly what cancels the variation.

Method I: The chapter instrument

1. Use $C = \pi - (A + B)$, expand $\cos C = -\cos(A + B)$, and collect terms in $\sin A, \cos A, \sin B, \cos B$.
2. Substituting the given restrictions and simplifying gives 1.

Method II: Independent check

1. Use the standard triangle identity obtained by applying the law of cosines to a unit circumcircle normalisation.
2. This route reaches the same value and rules out the trap, so the answer is 1.

INSIGHT

This is one of the safest tests for whether an angle-sum condition has actually been used.

№5 Sine Squares Minus the Cubic

●●●●● L4

EVALUATE

If $A + B + C = \pi$, simplify $\sin^2 A + \sin^2 B + \sin^2 C - 2 \cos A \cos B \cos C$.

sine-square cosine triangle

ANSWER

2

THE TRAP

The wrong answer 1 comes from using the cosine-square identity but forgetting that $\sin^2 = 1 - \cos^2$ changes three terms at once.

Method I: The chapter instrument

1. Replace $\sin^2$ by $1 - \cos^2$ and use $\cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C = 1$.
2. Substituting the given restrictions and simplifying gives 2.

Method II: Independent check

1. Equivalently expand $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C$ and subtract the cubic term.
2. This route reaches the same value and rules out the trap, so the answer is 2.

INSIGHT

The pair of identities for sine-squares and cosine-squares are dual, but the constants differ.

№6 Double-Angle Collapse

●●●●● L4

SIMPLIFY

If $A + B + C = \pi$, simplify $\sin 2A + \sin 2B + \sin 2C$.

double-angle product-to-sum triangle

ANSWER $4 \sin A \sin B \sin C$ **THE TRAP**

The tempting simplification 0 confuses $\sin(2A + 2B + 2C) = 0$ with the sum of the three sines.

Method I: The chapter instrument

1. Use $\sin 2A + \sin 2B = 2 \sin(A + B) \cos(A - B) = 2 \sin C \cos(A - B)$ and then add $\sin 2C$.
2. Substituting the given restrictions and simplifying gives $4 \sin A \sin B \sin C$.

Method II: Independent check

1. Write every term as $2 \sin A \cos A$ and use $\cos C = -\cos(A + B)$ to factor $4 \sin A \sin B \sin C$.
2. This route reaches the same value and rules out the trap, so the answer is $4 \sin A \sin B \sin C$.

INSIGHT

A zero total angle rarely means a zero sum; product-to-sum tells you what survives.

№7 Equilateral Product Ceiling

●●●●● L4

FIND THE MAXIMUM

For $A, B, C > 0$ with $A + B + C = \pi$, find the maximum of $\sin A \sin B \sin C$.

maximum

Jensen

triangle

MAXIMUM

$$\frac{3\sqrt{3}}{8}$$

THE TRAP

The seductive wrong answer is 1, obtained by maximising the three sines independently. The angle-sum constraint prevents all three from being $\pi/2$.

Method I: The chapter instrument

1. Apply Jensen to $\log \sin x$, which is concave on $(0, \pi)$; the product is maximised at $A = B = C = \pi/3$.
2. Substituting the given restrictions and simplifying gives $\frac{3\sqrt{3}}{8}$.

Method II: Independent check

1. Use $\sin A \sin B \leq \left(\frac{\sin A + \sin B}{2}\right)^2$ and smooth the angles until equality forces the equilateral case.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{3\sqrt{3}}{8}$.

INSIGHT

For positive triangle angles, symmetry plus strict concavity is often a complete extremum proof.

№8 Half-Tangent Sum Floor

●●●●● L4

FIND THE MINIMUM

If $A, B, C > 0$, $A + B + C = \pi$, find the minimum of $\tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2}$.

half-angle

minimum

uvw

MINIMUM

$$\sqrt{3}$$

THE TRAP

The trap is 0, because each half-angle tangent can be small separately. They cannot all be small when the half-angles sum to $\pi/2$.

Method I: The chapter instrument

1. Let x, y, z be the three half-angle tangents. Since $xy + yz + zx = 1$, $(x + y + z)^2 \geq 3(xy + yz + zx) = 3$.
2. Substituting the given restrictions and simplifying gives $\sqrt{3}$.

Method II: Independent check

1. By Jensen, $\tan x$ is convex on $(0, \pi/2)$; with half-angles summing to $\pi/2$, the sum is minimised at $\pi/6, \pi/6, \pi/6$.
2. This route reaches the same value and rules out the trap, so the answer is $\sqrt{3}$.

INSIGHT

The equality case again recognises the equilateral triangle.

№9 Cosine Sum Ceiling

●●●●● L4

FIND THE MAXIMUM

For $A, B, C > 0$ with $A + B + C = \pi$, find the maximum of $\cos A + \cos B + \cos C$.

maximum

cosine

triangle

MAXIMUM

$$\frac{3}{2}$$

THE TRAP

The false answer 3 maximises the cosines independently. The three angles cannot all be 0 under a positive sum of π .

Method I: The chapter instrument

1. Use $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} \leq 2 \cos \frac{\pi-C}{2}$ and then optimise in C .
2. Substituting the given restrictions and simplifying gives $\frac{3}{2}$.

Method II: Independent check

1. Use the identity $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$ and Euler's $r \leq R/2$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{3}{2}$.

INSIGHT

The trig and geometry proofs meet at the same equality case: $A = B = C = \pi/3$.

№10 A Determinant That Refuses to Vary

●●●●● L5

EVALUATE

$$\text{If } A + B + C = \pi, \text{ evaluate } \begin{vmatrix} 1 & \cos A & \sin A \\ 1 & \cos B & \sin B \\ 1 & \cos C & \sin C \end{vmatrix}^2.$$

determinant

half-angle

vandermonde

ANSWER

$$16 \sin^2 \frac{A-B}{2} \sin^2 \frac{B-C}{2} \sin^2 \frac{C-A}{2}$$

THE TRAP

The attractive wrong answer is 0, because the first column is constant. The rows are not linearly dependent unless two angles coincide modulo 2π .

Method I: The chapter instrument

1. Subtract the first row from the other two and use product-to-sum on cosine and sine differences; the half-angle factors appear immediately.
2. Substituting the given restrictions and simplifying gives $16 \sin^2 \frac{A-B}{2} \sin^2 \frac{B-C}{2} \sin^2 \frac{C-A}{2}$.

Method II: Independent check

1. Interpret $(\cos A, \sin A)$ as points on the unit circle: the determinant is twice the signed area of the triangle formed by those points.
2. This route reaches the same value and rules out the trap, so the answer is $16 \sin^2 \frac{A-B}{2} \sin^2 \frac{B-C}{2} \sin^2 \frac{C-A}{2}$.

INSIGHT

Circle geometry often turns a trigonometric determinant into an area formula.

CHAPTER II

Product-to-Sum Telescopes



Product-to-sum and adjacent differences leave only endpoints.

№11 Cotangent Difference Chain

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \frac{1}{\sin kx \sin(k+1)x}$$

telescoping cotangent finite-sum

ANSWER

$$\frac{\cot x - \cot(n+1)x}{\sin x}$$

THE TRAP

The trap is $\frac{n}{\sin^2 x}$, treating all sine factors as if $kx = x$. The adjacent angles are the whole point.

Method I: The chapter instrument

1. Use $\cot u - \cot v = \frac{\sin(v-u)}{\sin u \sin v}$ with $u = kx, v = (k+1)x$.
2. Substituting the given restrictions and simplifying gives $\frac{\cot x - \cot(n+1)x}{\sin x}$.

Method II: Independent check

1. Multiply the proposed answer by $\sin x$ and expand it as a telescoping sum of cotangent differences.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\cot x - \cot(n+1)x}{\sin x}$.

INSIGHT

Whenever neighbouring sine factors occur, look for a cotangent difference.

№12 Dyadic Cosine Product

●●●●● L3

EVALUATE

$$\prod_{k=0}^{n-1} \cos(2^k x)$$

dyadic product double-angle

ANSWER

$$\frac{\sin(2^n x)}{2^n \sin x}$$

THE TRAP

The wrong answer $\cos((2^n - 1)x)$ adds angles inside cosines. Products require repeated double-angle formulas, not angle addition.

Method I: The chapter instrument

1. Repeatedly apply $2 \sin y \cos y = \sin 2y$, starting with $y = x$.
2. Substituting the given restrictions and simplifying gives $\frac{\sin(2^n x)}{2^n \sin x}$.

Method II: Independent check

1. Induct on n : multiplying the n th formula by $\cos(2^n x)$ produces the $(n+1)$ st formula.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\sin(2^n x)}{2^n \sin x}$.

INSIGHT

This is the standard dyadic product behind many disguised telescope problems.

№13 Sine Arithmetic Progression

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \sin kx$$

finite-sum

product-to-sum

progression

ANSWER

$$\frac{\sin \frac{nx}{2} \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$$

THE TRAP

The trap is $n \sin x$, valid only when every term has the same angle. The phase drift contributes the second sine factor.

Method I: The chapter instrument

1. Multiply the sum by $2 \sin(x/2)$ and use $2 \sin(kx) \sin(x/2) = \cos((k - \frac{1}{2})x) - \cos((k + \frac{1}{2})x)$.
2. Substituting the given restrictions and simplifying gives $\frac{\sin \frac{nx}{2} \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$.

Method II: Independent check

1. Sum the imaginary parts of $e^{ix} + e^{2ix} + \dots + e^{inx}$ and simplify the geometric series.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\sin \frac{nx}{2} \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$.

INSIGHT

Arithmetic progressions of angles are geometric progressions on the unit circle.

№14 Cosine Arithmetic Progression

●●●●● L3

EVALUATE

$$\sum_{k=1}^n \cos kx$$

finite-sum

cosine

progression

ANSWER

$$\frac{\sin \frac{nx}{2} \cos \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$$

THE TRAP

The wrong answer $n \cos x$ ignores the rotating phase. The denominator $\sin(x/2)$ records the spacing.

Method I: The chapter instrument

1. Multiply by $2 \sin(x/2)$ and use $2 \cos(kx) \sin(x/2) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x)$.
2. Substituting the given restrictions and simplifying gives $\frac{\sin \frac{nx}{2} \cos \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$.

Method II: Independent check

1. Take real parts in the geometric series $\sum_{k=1}^n e^{ikx}$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\sin \frac{\pi x}{2} \cos \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$.

INSIGHT

The sine and cosine progression formulas are the same telescope seen through different parts.

№15 Arctangent Ladder

••••• L4

EVALUATE

$$\sum_{k=1}^n \tan^{-1} \frac{1}{k^2 + k + 1}$$

arctangent

telescope

branch

ANSWER

$$\tan^{-1}(n+1) - \frac{\pi}{4}$$

THE TRAP

The trap is $\tan^{-1} \sum \frac{1}{k^2+k+1}$. Arctangent does not distribute over addition; the difference formula is what telescopes.

Method I: The chapter instrument

1. Use $\tan^{-1}(k+1) - \tan^{-1}k = \tan^{-1} \frac{1}{k^2+k+1}$, with all terms positive so no branch correction appears.
2. Substituting the given restrictions and simplifying gives $\tan^{-1}(n+1) - \frac{\pi}{4}$.

Method II: Independent check

1. Take tangent of partial sums after writing each term as an angle between slopes k and $k+1$.
2. This route reaches the same value and rules out the trap, so the answer is $\tan^{-1}(n+1) - \frac{\pi}{4}$.

INSIGHT

Even inverse-trig telescopes work only after checking the branch.

№16 Sine Ratio Collapse

••••• L3

SIMPLIFY

$$\prod_{k=1}^n \frac{\sin(2^k x)}{\sin(2^{k-1} x)}$$

product

dyadic

cancellation

ANSWER

$$\frac{\sin(2^n x)}{\sin x}$$

THE TRAP

The tempting answer 2^n comes from replacing each sine ratio by an angle ratio, which is only a limiting approximation near zero.

Method I: The chapter instrument

1. Write out the first few factors; every intermediate sine cancels.
2. Substituting the given restrictions and simplifying gives $\frac{\sin(2^n x)}{\sin x}$.

Method II: Independent check

1. Use the dyadic cosine product formula and multiply both sides by $2^n \prod \cos(2^{k-1} x)$ to see the same endpoint ratio.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\sin(2^n x)}{\sin x}$.

INSIGHT

Before using identities, always check whether a product is already telescoping by cancellation.

№17 Secant Neighbour Sum

••••• L4

EVALUATE

$$\sum_{k=1}^n \frac{1}{\cos kx \cos(k+1)x}$$

tangent secant finite-sum

ANSWER

$$\frac{\tan((n+1)x) - \tan x}{\sin x}$$

THE TRAP

The trap is $\frac{n}{\cos^2 x}$, flattening all neighbouring angles into one. The tangent difference keeps the endpoints.

Method I: The chapter instrument

1. Use $\tan v - \tan u = \frac{\sin(v-u)}{\cos u \cos v}$ with $u = kx, v = (k+1)x$.
2. Substituting the given restrictions and simplifying gives $\frac{\tan((n+1)x) - \tan x}{\sin x}$.

Method II: Independent check

1. Differentiate the logarithm of a cosine product formally and compare adjacent tangent terms.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\tan((n+1)x) - \tan x}{\sin x}$.

INSIGHT

Sine-neighbour sums telescope with cotangent; cosine-neighbour sums telescope with tangent.

№18 Odd Sine Block

••••• L3

EVALUATE

$$\sum_{k=1}^n \sin((2k-1)x)$$

odd-angles finite-sum product-to-sum

ANSWER

$$\frac{\sin^2(nx)}{\sin x}$$

THE TRAP

The wrong answer 0 appears if one imagines odd sines pairing symmetrically. The block starts at x and ends at $(2n-1)x$, so the endpoints do not cancel.

Method I: The chapter instrument

1. Multiply by $2 \sin x$ and use $2 \sin((2k-1)x) \sin x = \cos((2k-2)x) - \cos(2kx)$.
2. Substituting the given restrictions and simplifying gives $\frac{\sin^2(nx)}{\sin x}$.

Method II: Independent check

1. Sum the imaginary parts of $e^{ix}(1 + e^{2ix} + \dots + e^{2(n-1)ix})$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{\sin^2(nx)}{\sin x}$.

INSIGHT

Odd-angle progressions are just ordinary progressions with step $2x$.

№19 Boundary Cosines Survive

●●●●● L4

PROVE THAT

$$2 \sin \frac{x}{2} \sum_{k=1}^n \sin kx = \cos \frac{x}{2} - \cos \frac{(2n+1)x}{2}$$

proof finite-sum boundary

CONCLUSION

$$\cos \frac{x}{2} - \cos \frac{(2n+1)x}{2}$$

THE TRAP

The seductive wrong conclusion is 0, from expecting all cosines to cancel. Telescopes keep the first and last boundary terms.

Method I: The chapter instrument

1. Use $2 \sin(kx) \sin(x/2) = \cos((k - \frac{1}{2})x) - \cos((k + \frac{1}{2})x)$ and sum over k .
2. Substituting the given restrictions and simplifying gives $\cos \frac{x}{2} - \cos \frac{(2n+1)x}{2}$.

Method II: Independent check

1. Insert the closed form for $\sum \sin kx$ and use $2 \sin u \sin v = \cos(u - v) - \cos(u + v)$.
2. This route reaches the same value and rules out the trap, so the answer is $\cos \frac{x}{2} - \cos \frac{(2n+1)x}{2}$.

INSIGHT

A telescope is never empty; its boundary terms are the answer.

№20 Zeros of a Dyadic Product

●●●●● L4

SOLVE

$$\text{For } m \geq 1, \text{ solve } \prod_{k=0}^{m-1} 2 \cos(2^k x) = 0 \text{ on } [0, \pi).$$

zero-counting dyadic equation

SOLUTIONS

$$\left\{ \frac{j\pi}{2^m} : 1 \leq j \leq 2^m - 1 \right\}$$

THE TRAP

The trap is to list only $x = \pi/2$ from the first cosine. Later factors create all dyadic subdivision points.

Method I: The chapter instrument

1. Use $\prod_{k=0}^{m-1} 2 \cos(2^k x) = \frac{\sin(2^m x)}{\sin x}$ and exclude the endpoint where the quotient form was divided by $\sin x$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{j\pi}{2^m} : 1 \leq j \leq 2^m - 1 \right\}$.

Method II: Independent check

1. Alternatively solve each factor $\cos(2^k x) = 0$ and merge the dyadic fractions into the single set $j\pi/2^m$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{j\pi}{2^m} : 1 \leq j \leq 2^m - 1 \right\}$.

INSIGHT

The product identity is a compact way of removing duplicate zeros.

CHAPTER III

Unit Roots & Finite Trig Sums

Finite sums and products answered by the unit circle.



№21 The Vanishing Polygon

●●●●● L3

EVALUATE

$$\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} \quad (n > 1)$$

roots-of-unity

cosine-sum

polygon

ANSWER

0

THE TRAP

The trap is n , treating every cosine as the first term 1. The remaining vertices of the regular polygon cancel it exactly.

Method I: The chapter instrument

1. Take real parts of $1 + \omega + \dots + \omega^{n-1} = 0$ for $\omega = e^{2\pi i/n}$.
2. Substituting the given restrictions and simplifying gives 0.

Method II: Independent check

1. Pair vertices of the regular n -gon by symmetry; their horizontal components sum to zero.
2. This route reaches the same value and rules out the trap, so the answer is 0.

INSIGHT

Most finite trigonometric sums are shadows of a roots-of-unity sum.

№22 Sine Product of the Regular Polygon

●●●●● L4

EVALUATE

$$\prod_{k=1}^{n-1} \sin \frac{\pi k}{n}$$

sine-product

roots-of-unity

polynomial

ANSWER

$\frac{n}{2^{n-1}}$

THE TRAP

The common wrong answer 1 comes from pairing $\sin(\pi k/n)$ with $\sin(\pi - \pi k/n)$ and forgetting their actual magnitudes.

Method I: The chapter instrument

1. Evaluate $\frac{z^n - 1}{z - 1}$ at $z = 1$ after factoring $z^n - 1$ over the nontrivial roots and taking moduli.
2. Substituting the given restrictions and simplifying gives $\frac{n}{2^{n-1}}$.

Method II: Independent check

1. Use $|1 - e^{2\pi i k/n}| = 2 \sin(\pi k/n)$ and multiply over $k = 1, \dots, n-1$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n}{2^{n-1}}$.

INSIGHT

A product of chord lengths in a regular polygon is usually a disguised derivative at 1.

№23 Second Moment of Cosines

●●●●● L3

EVALUATE

$$\sum_{k=0}^{n-1} \cos^2 \frac{2\pi k}{n} \quad (n > 2)$$

power-sum cosine orthogonality

ANSWER

$$\frac{n}{2}$$

THE TRAP

The trap is n , replacing every square by 1. The oscillating $\cos(4\pi k/n)$ part vanishes.

Method I: The chapter instrument

1. Use $\cos^2 \theta = (1 + \cos 2\theta)/2$ and the vanishing of the nonzero root sum.
2. Substituting the given restrictions and simplifying gives $\frac{n}{2}$.

Method II: Independent check

1. Interpret the sum as the squared x -coordinates of the vertices; rotational symmetry gives half the total squared radius sum.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n}{2}$.

INSIGHT

Power sums become easy after reducing powers to multiple angles.

№24 Fourth Moment Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^{n-1} \cos^4 \frac{2\pi k}{n} \quad (n \geq 5)$$

power-sum fourth-power filter

ANSWER

$$\frac{3n}{8}$$

THE TRAP

The attractive answer $n/2$ copies the square-moment result. The fourth power has a different constant term.

Method I: The chapter instrument

1. Use $\cos^4 \theta = \frac{3+4\cos 2\theta+\cos 4\theta}{8}$; both nonconstant sums vanish for $n \geq 5$.
2. Substituting the given restrictions and simplifying gives $\frac{3n}{8}$.

Method II: Independent check

1. Write cosines with ω^k and keep only exponents divisible by n ; only the constant exponent survives.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{3n}{8}$.

INSIGHT

The condition $n \geq 5$ is not decoration: small n can let the fourth harmonic survive.

№25 Reciprocal Sine Squares

●●●●● L5

EVALUATE

$$\sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n}$$

cosecant

derivative

roots

ANSWER

$$\frac{n^2 - 1}{3}$$

THE TRAP

The trap is $n - 1$, pretending every sine equals 1. Near 0 the reciprocal terms are large and dominate.

Method I: The chapter instrument

1. Differentiate the logarithmic derivative of $\frac{z^n - 1}{z - 1}$ and evaluate the second-order term at $z = 1$.
2. Substituting the given restrictions and simplifying gives $\frac{n^2 - 1}{3}$.

Method II: Independent check

1. Use the partial fraction expansion of $\cot nx$ in terms of $\cot(x + k\pi/n)$, then compare the coefficient of x near 0.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n^2 - 1}{3}$.

INSIGHT

Second derivatives of root polynomials are the clean way to sum reciprocal squares.

№26 Weighted Sine Sum

●●●●● L4

EVALUATE

$$\sum_{k=1}^{n-1} k \sin \frac{2\pi k}{n}$$

weighted-sum

roots-of-unity

imaginary-part

ANSWER

$$-\frac{n}{2} \cot \frac{\pi}{n}$$

THE TRAP

The wrong answer 0 pairs k and $n - k$ but misses that the weights are not equal.

Method I: The chapter instrument

1. For $\omega = e^{2\pi i/n}$, use $\sum_{k=1}^{n-1} k\omega^k = -\frac{n}{1-\omega}$ and take imaginary parts.
2. Substituting the given restrictions and simplifying gives $-\frac{n}{2} \cot \frac{\pi}{n}$.

Method II: Independent check

1. Pair k with $n - k$: the sine values cancel in sign but leave coefficient $k - (n - k)$, producing a cotangent sum.
2. This route reaches the same value and rules out the trap, so the answer is $-\frac{n}{2} \cot \frac{\pi}{n}$.

INSIGHT

Weights break symmetry; roots of unity keep track of the lost balance.

№27 Cosine Gap Product

••••• L4

EVALUATE

$$\prod_{k=1}^{n-1} \left(1 - \cos \frac{2\pi k}{n} \right)$$

product cosine roots

ANSWER

$$\frac{n^2}{2^{n-1}}$$

THE TRAP

The trap is 0, because $1 - \cos 0 = 0$ looks nearby. The product deliberately omits $k = 0$.

Method I: The chapter instrument

1. Use $1 - \cos 2u = 2 \sin^2 u$ and the sine-product formula.
2. Substituting the given restrictions and simplifying gives $\frac{n^2}{2^{n-1}}$.

Method II: Independent check

1. Write $1 - \cos \theta = \frac{1}{2} |1 - e^{i\theta}|^2$ and multiply the chord lengths from 1 to all other roots.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n^2}{2^{n-1}}$.

INSIGHT

Cosine gap products are squared chord products in disguise.

№28 Odd Half-Step Energy

••••• L3

EVALUATE

$$\sum_{k=0}^{n-1} \sin^2 \frac{(2k+1)\pi}{2n}$$

half-step sine-square symmetry

ANSWER

$$\frac{n}{2}$$

THE TRAP

The wrong answer n assumes every odd half-step hits a maximum. Only one does; symmetry averages the squares to $1/2$.

Method I: The chapter instrument

1. Use $\sin^2 \theta = (1 - \cos 2\theta)/2$; the remaining cosine sum is over the n roots of -1 and vanishes.
2. Substituting the given restrictions and simplifying gives $\frac{n}{2}$.

Method II: Independent check

1. Reflect the n points across the axes; their squared vertical coordinates share total energy equally with squared horizontal coordinates.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n}{2}$.

INSIGHT

Half-step grids behave like roots of $z^n = -1$.

№29 Squared Chord Product

••••• L4

EVALUATE

$$\prod_{k=1}^{n-1} \left(2 - 2 \cos \frac{2\pi k}{n} \right)$$

chords product unit-circle

ANSWER

$$n^2$$

THE TRAP

The tempting answer n forgets that each factor is a squared chord length, not the chord length itself.

Method I: The chapter instrument

1. Since $2 - 2 \cos \theta = |1 - e^{i\theta}|^2$, the product is the square of $\prod_{k=1}^{n-1} |1 - \omega^k| = n$.
2. Substituting the given restrictions and simplifying gives n^2 .

Method II: Independent check

1. Use $2 - 2 \cos 2u = 4 \sin^2 u$ and the sine-product formula.
2. This route reaches the same value and rules out the trap, so the answer is n^2 .

INSIGHT

Taking moduli often squares a product; watch that exponent.

№30 Frequency Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^{n-1} \cos \frac{2\pi mk}{n}$$

filter orthogonality frequency

ANSWER

$$\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

THE TRAP

The trap is 0 always, overlooking the resonant case when the frequency is a multiple of n and every term is 1.

Method I: The chapter instrument

1. Let $\omega = e^{2\pi i/n}$. The complex sum is $\sum (\omega^m)^k$, which is n if $\omega^m = 1$ and 0 otherwise.
2. Substituting the given restrictions and simplifying gives $\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$

Method II: Independent check

1. View the terms as projections of a regular polygon traced with step m ; it closes symmetrically unless every point is 1.
2. This route reaches the same value and rules out the trap, so the answer is $\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$

INSIGHT

This is the finite Fourier filter in its most elementary form.

CHAPTER IV

Chebyshev & Multiple Angles

T_n

Multiple-angle polynomials turn trigonometry into algebra.

№31 The Fifth-Angle Polynomial

●●●●● L3

SIMPLIFY

$\cos 5x$ in powers of $c = \cos x$

multiple-angle polynomial Chebyshev

ANSWER

$$16c^5 - 20c^3 + 5c$$

THE TRAP

The trap is $5 \cos x$, linearising a highly nonlinear multiple-angle expression.

Method I: The chapter instrument

1. Use $\cos(2x)$ and $\cos(3x)$, or the recurrence $T_{r+1}(c) = 2cT_r(c) - T_{r-1}(c)$ up to T_5 .
2. Substituting the given restrictions and simplifying gives $16c^5 - 20c^3 + 5c$.

Method II: Independent check

1. Take the real part of $(c + i\sqrt{1-c^2})^5$ and collect powers of c .
2. This route reaches the same value and rules out the trap, so the answer is $16c^5 - 20c^3 + 5c$.

INSIGHT

Chebyshev polynomials are multiple-angle formulas with the trigonometry removed.

№32 Three Roots of a Cubic Cosine

●●●●● L4

SOLVE

$$8x^3 - 6x = 1$$

cubic triple-angle roots

SOLUTIONS

$$\left\{ \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9} \right\}$$

THE TRAP

The trap is keeping only $x = \cos(\pi/9)$. A cubic has three real roots here; cosine periodicity supplies all of them.

Method I: The chapter instrument

1. Put $x = \cos \theta$. Then $8x^3 - 6x = 2 \cos 3\theta$, so $\cos 3\theta = 1/2$.
2. Substituting the given restrictions and simplifying gives $\left\{ \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9} \right\}$.

Method II: Independent check

1. Solve $3\theta = \pm\pi/3 \pmod{2\pi}$ and take the three distinct cosine values.
2. This route reaches the same value and rules out the trap, so the answer is $\{\cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}\}$.

INSIGHT

Trigonometric solution of cubics is not a trick; it is Chebyshev inversion.

№33 The Heptagon Cubic

••••• L5

PROVE THAT

$$\alpha = 2 \cos \frac{\pi}{7} \text{ satisfies } \alpha^3 - \alpha^2 - 2\alpha + 1 = 0.$$

minimal-polynomial

heptagon

Chebyshev

CONCLUSION

$$\alpha^3 - \alpha^2 - 2\alpha + 1 = 0$$

THE TRAP

The trap is the nearby but false $\alpha^3 - 3\alpha - 1 = 0$, which belongs to a different angle.

Method I: The chapter instrument

1. Use $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ with $\theta = \pi/7$, together with $\cos 3\theta = \cos(\pi - 4\theta) = -\cos 4\theta$.
2. Substituting the given restrictions and simplifying gives $\alpha^3 - \alpha^2 - 2\alpha + 1 = 0$.

Method II: Independent check

1. Use $\zeta = e^{i\pi/7}$ and the relation $1 + \zeta^2 + \zeta^4 + \zeta^6 + \zeta^8 + \zeta^{10} + \zeta^{12} = 0$, then group reciprocal powers.
2. This route reaches the same value and rules out the trap, so the answer is $\alpha^3 - \alpha^2 - 2\alpha + 1 = 0$.

INSIGHT

Regular polygons hide algebraic equations of surprisingly low degree.

№34 Nested Radical Angle

••••• L4

EVALUATE

$$\sqrt{2 + \sqrt{2 + \sqrt{2}}}$$

nested-radical

half-angle

cosine

ANSWER

$$2 \cos \frac{\pi}{16}$$

THE TRAP

The trap is 2, since the radicals look like they are converging. This finite radical is close to but not equal to 2.

Method I: The chapter instrument

1. Start with $\sqrt{2} = 2 \cos(\pi/4)$ and repeatedly use $\sqrt{2 + 2 \cos u} = 2 \cos(u/2)$.
2. Substituting the given restrictions and simplifying gives $2 \cos \frac{\pi}{16}$.

Method II: Independent check

1. Square the claimed value successively using $2 \cos^2 u = 1 + \cos 2u$ until the inner radical becomes $\sqrt{2}$.
2. This route reaches the same value and rules out the trap, so the answer is $2 \cos \frac{\pi}{16}$.

INSIGHT

Finite nested radicals remember exactly how many half-angle steps were taken.

№35 Sine Five Equals Sine One

●●●●● L4

SOLVE

$$\sin 5x = \sin x \quad \text{for } 0 \leq x < 2\pi$$

equation multiple-angle root-counting

SOLUTIONS

$$\left\{ 0, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6} \right\}$$

THE TRAP

The trap is listing six roots from $5x = \pi - x$ and forgetting the branch $5x = x$; another trap is double-counting their overlap.

Method I: The chapter instrument

1. Use $\sin u = \sin v$: either $u = v + 2k\pi$ or $u = \pi - v + 2k\pi$, then merge duplicate roots.
2. Substituting the given restrictions and simplifying gives $\left\{ 0, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6} \right\}$.

Method II: Independent check

1. Factor $\sin 5x - \sin x = 2 \cos 3x \sin 2x$ and solve the two simpler equations on the interval.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ 0, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6} \right\}$.

INSIGHT

Product-to-sum is often the cleanest root-counting form of a multiple-angle equation.

№36 Sine Quotient Polynomial

●●●●● L4

SIMPLIFY

$$\frac{\sin nx}{\sin x}$$

Chebyshev-U sine-quotient recurrence

ANSWER

$$U_{n-1}(\cos x)$$

THE TRAP

The wrong answer n is only the limit as $x \rightarrow 0$, not the identity for general x .

Method I: The chapter instrument

1. Both sides satisfy $S_{n+1} = 2 \cos x S_n - S_{n-1}$ with initial values $S_1 = 1, S_2 = 2 \cos x$.
2. Substituting the given restrictions and simplifying gives $U_{n-1}(\cos x)$.

Method II: Independent check

1. Use e^{ix} : $(e^{inx} - e^{-inx}) / (e^{ix} - e^{-ix})$ is a symmetric polynomial in $e^{ix} + e^{-ix} = 2 \cos x$.
2. This route reaches the same value and rules out the trap, so the answer is $U_{n-1}(\cos x)$.

INSIGHT

The second-kind Chebyshev polynomial is the natural language of sine quotients.

№37 Zeros of T_n

●●●●● L4

EVALUATE

$$\prod_{k=1}^n \left(x - \cos \frac{(2k-1)\pi}{2n} \right)$$

zeros Chebyshev product

ANSWER

$$2^{1-n} T_n(x)$$

THE TRAP

The trap is missing the leading factor and writing $T_n(x)$. Chebyshev polynomials are not monic.

Method I: The chapter instrument

1. The displayed cosine values are exactly the zeros of $T_n(x) = \cos(n \cos^{-1} x)$.
2. Substituting the given restrictions and simplifying gives $2^{1-n} T_n(x)$.

Method II: Independent check

1. Compare leading coefficients: T_n has leading coefficient 2^{n-1} for $n \geq 1$.
2. This route reaches the same value and rules out the trap, so the answer is $2^{1-n} T_n(x)$.

INSIGHT

Root lists determine a polynomial only up to its leading coefficient.

№38 Binomial Cosine Filter

●●●●● L4

EVALUATE

$$\sum_{k=0}^n \binom{n}{k} \cos((n-2k)x)$$

binomial complex-numbers cosine

ANSWER

$$2^n \cos^n x$$

THE TRAP

The tempting answer 0 comes from expecting the cosine terms to cancel. The binomial weights destroy that cancellation.

Method I: The chapter instrument

1. Take real parts of $\sum_{k=0}^n \binom{n}{k} e^{i(n-2k)x} = e^{inx} (1 + e^{-2ix})^n$.
2. Substituting the given restrictions and simplifying gives $2^n \cos^n x$.

Method II: Independent check

1. Rewrite $e^{ix} + e^{-ix} = 2 \cos x$ and observe that the whole binomial sum is already real.
2. This route reaches the same value and rules out the trap, so the answer is $2^n \cos^n x$.

INSIGHT

Binomial coefficients often signal a disguised power of $2 \cos x$.

№39 Golden Cosine Consequence

●●●●● L4

EVALUATE

If $2 \cos x$ satisfies $t^2 - t - 1 = 0$, find $\cos 5x$.

golden-ratio pentagon multiple-angle

ANSWER

$$-1$$

THE TRAP

The trap is 1, from remembering fifth roots of unity but choosing the wrong pentagon angle.

Method I: The chapter instrument

1. The roots give $\cos x = (1 \pm \sqrt{5})/4$, i.e. $x \equiv \pm\pi/5$ or $\pm 3\pi/5 \pmod{2\pi}$.
2. Substituting the given restrictions and simplifying gives -1 .

Method II: Independent check

1. Substitute $c = \cos x$ into $16c^5 - 20c^3 + 5c$ and reduce using $4c^2 - 2c - 1 = 0$.
2. This route reaches the same value and rules out the trap, so the answer is -1 .

INSIGHT

The pentagon quadratic remembers that five times the angle lands at an odd multiple of π .

No. 40 Polynomial Inequality Translation

••••• L5

PROVE THAT

$$\cos 4x \geq 8 \cos^4 x - 8 \cos^2 x \quad \text{for all real } x.$$

inequality

multiple-angle

polynomial

CONCLUSION

$$\cos 4x - (8 \cos^4 x - 8 \cos^2 x) = 1$$

THE TRAP

The trap is to test a few angles and call it an inequality. It is actually an identity with a constant gap.

Method I: The chapter instrument

1. Use $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$.
2. Substituting the given restrictions and simplifying gives $\cos 4x - (8 \cos^4 x - 8 \cos^2 x) = 1$.

Method II: Independent check

1. Apply the Chebyshev recurrence to get $T_4(c) = 8c^4 - 8c^2 + 1$.
2. This route reaches the same value and rules out the trap, so the answer is $\cos 4x - (8 \cos^4 x - 8 \cos^2 x) = 1$.

INSIGHT

Some inequalities are identities wearing a missing constant.

CHAPTER V

Trigonometric Substitution $x = \tan \theta$ Engines

The right parameter makes radicals and ranges linear.

№41 Cosine From a Rational Parameter

●●●●● L3

FIND THE RANGE

$$\frac{1-x^2}{1+x^2} \quad (x \in \mathbb{R})$$

range

tangent-half-angle

rational

RANGE

$$(-1, 1]$$

THE TRAP

The trap is $(-1, 1)$, because endpoints look unreachable. But $x = 0$ gives 1, and the limit as $|x| \rightarrow \infty$ gives -1 as an attained endpoint only in the extended parameter? Here the expression approaches -1 but never equals it for finite x ; the exact range is $(-1, 1]$.

Method I: The chapter instrument

1. Set $x = \tan(\theta/2)$ to get $\frac{1-x^2}{1+x^2} = \cos \theta$; finite real x corresponds to $\theta \not\equiv \pi \pmod{2\pi}$.
2. Substituting the given restrictions and simplifying gives $[-1, 1]$.

Method II: Independent check

1. Solve $y = (1-x^2)/(1+x^2)$: $x^2 = (1-y)/(1+y)$, so $-1 < y \leq 1$.
2. This route reaches the same value and rules out the trap, so the answer is $[-1, 1]$.

INSIGHT

The parametrisation shows the natural bound; the algebra decides whether the endpoint is attained.

№42 The Unit Circle Parametrisation

●●●●● L3

PROVE THAT

$$\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{1-t^2}{1+t^2}\right)^2 = 1$$

parametrisation

unit-circle

identity

CONCLUSION

$$1$$

THE TRAP

The tempting wrong expression is $\frac{1+t^4}{(1+t^2)^2}$, from dropping the middle terms during expansion.

Method I: The chapter instrument

1. Expand the numerator: $4t^2 + (1 - 2t^2 + t^4) = (1 + t^2)^2$.

2. Substituting the given restrictions and simplifying gives 1.

Method II: Independent check

1. Set $t = \tan(\theta/2)$; the two fractions become $\sin \theta$ and $\cos \theta$.
2. This route reaches the same value and rules out the trap, so the answer is 1.

INSIGHT

This is the rational parametrisation of the unit circle.

No 43 A Rational Sine Equation

●●●●● L4

SOLVE

$$\frac{2x}{1+x^2} = a \quad (x \in \mathbb{R})$$

parameter quadratic range

SOLUTIONS

$$\begin{cases} \left\{ \frac{1 \pm \sqrt{1-a^2}}{a} \right\}, & 0 < |a| \leq 1, \\ \{0\}, & a = 0, \\ \emptyset, & |a| > 1. \end{cases}$$

THE TRAP

The trap is solving the quadratic and accepting complex-looking or division-by-zero cases blindly. The parameter value $a = 0$ must be split off.

Method I: The chapter instrument

1. Recognise $2x/(1+x^2) = \sin \theta$ under $x = \tan(\theta/2)$, so real solutions require $|a| \leq 1$.

2. Substituting the given restrictions and simplifying gives $\begin{cases} \left\{ \frac{1 \pm \sqrt{1-a^2}}{a} \right\}, & 0 < |a| \leq 1, \\ \{0\}, & a = 0, \\ \emptyset, & |a| > 1. \end{cases}$.

Method II: Independent check

1. Rearrange to $ax^2 - 2x + a = 0$; its discriminant is $4(1-a^2)$, with $a = 0$ handled separately.

2. This route reaches the same value and rules out the trap, so the answer is $\begin{cases} \left\{ \frac{1 \pm \sqrt{1-a^2}}{a} \right\}, & 0 < |a| \leq 1, \\ \{0\}, & a = 0, \\ \emptyset, & |a| > 1. \end{cases}$.

INSIGHT

A trig substitution gives the domain of the parameter before algebra gives the roots.

No 44 Linear Over Quadratic Peak

●●●●● L4

FIND THE RANGE

$$\frac{3x+4}{x^2+1} \quad (x \in \mathbb{R})$$

range discriminant substitution

RANGE

$$\left[-\frac{1}{2}, \frac{9}{2} \right]$$

THE TRAP

The trap is $[-5, 5]$, using $3 \sin \theta + 4 \cos \theta$ but forgetting the half-angle factor changes the constant term.

Method I: The chapter instrument

1. Set the expression equal to y ; the quadratic $yx^2 - 3x + (y - 4) = 0$ must have nonnegative discriminant.
2. Substituting the given restrictions and simplifying gives $[-\frac{1}{2}, \frac{9}{2}]$.

Method II: Independent check

1. With $x = \tan(\theta/2)$, rewrite the expression as $\frac{3}{2}\sin\theta + 2(1 + \cos\theta)$ and optimise the resulting sinusoid with a constant.
2. This route reaches the same value and rules out the trap, so the answer is $[-\frac{1}{2}, \frac{9}{2}]$.

INSIGHT

For rational functions of degree two, the discriminant is often the fastest range machine.

No45 Twelve-One Peak

●●●●● L4

FIND THE RANGE

$$\frac{5 + 12x}{1 + x^2} \quad (x \in \mathbb{R})$$

range quadratic extremum

RANGE

$$[-4, 9]$$

THE TRAP

The trap is $[-13, 13]$, treating the numerator as $5\cos\theta + 12\sin\theta$ without accounting for the denominator's half-angle form.

Method I: The chapter instrument

1. Set $y = (5 + 12x)/(1 + x^2)$ and require $yx^2 - 12x + (y - 5) = 0$ to have a real root.
2. Substituting the given restrictions and simplifying gives $[-4, 9]$.

Method II: Independent check

1. Use $x = \tan(\theta/2)$ to obtain $\frac{5}{2}(1 + \cos\theta) + 6\sin\theta$ and optimise the shifted sinusoid.
2. This route reaches the same value and rules out the trap, so the answer is $[-4, 9]$.

INSIGHT

A constant plus a sinusoid has a shifted range, not a centred one.

No46 Unit Circle From a Line

●●●●● L3

SIMPLIFY

$$\sin(2 \tan^{-1} x)$$

inverse-tangent double-angle rational

ANSWER

$$\frac{2x}{1 + x^2}$$

THE TRAP

The tempting answer $2x$ uses $\sin 2u = 2\sin u$ instead of $2\sin u \cos u$.

Method I: The chapter instrument

1. Let $u = \tan^{-1} x$, so $\tan u = x$; draw a right triangle with opposite x and adjacent 1.
2. Substituting the given restrictions and simplifying gives $\frac{2x}{1+x^2}$.

Method II: Independent check

1. Use $\sin 2u = \frac{2 \tan u}{1 + \tan^2 u}$ directly.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{2x}{1+x^2}$.

INSIGHT

The formulas for $\sin(2 \tan^{-1} x)$ and $\cos(2 \tan^{-1} x)$ are the tangent half-angle map in reverse.

No. 47 Cosine From Inverse Tangent

● ● ● ● ● L3

SIMPLIFY

$$\cos(2 \tan^{-1} x)$$

inverse-tangent

double-angle

rational

ANSWER

$$\frac{1-x^2}{1+x^2}$$

THE TRAP

The trap is $1 - 2x^2$, using a small-angle approximation as an identity.

Method I: The chapter instrument

1. Let $u = \tan^{-1} x$ and use $\cos 2u = \frac{1 - \tan^2 u}{1 + \tan^2 u}$.
2. Substituting the given restrictions and simplifying gives $\frac{1-x^2}{1+x^2}$.

Method II: Independent check

1. From the right triangle, $\sin u = x/\sqrt{1+x^2}$ and $\cos u = 1/\sqrt{1+x^2}$; subtract squares.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{1-x^2}{1+x^2}$.

INSIGHT

Inverse tangent turns rational functions into ordinary double-angle identities.

No. 48 Radical Touches the Line

● ● ● ● ● L4

SOLVE

$$\sqrt{1-x^2} + x = 1 \quad (-1 \leq x \leq 1)$$

radical

sine-substitution

extraneous

SOLUTIONS

$$\{0, 1\}$$

THE TRAP

The trap is accepting all roots of the squared equation without checking the original square root condition.

Method I: The chapter instrument

1. Put $x = \sin \theta$ with $\theta \in [-\pi/2, \pi/2]$, so the equation becomes $\cos \theta + \sin \theta = 1$.
2. Substituting the given restrictions and simplifying gives $\{0, 1\}$.

Method II: Independent check

1. Square only after noting $1 - x \geq 0$; $1 - x^2 = (1 - x)^2$ gives $x = 0$ or $x = 1$, both valid.
2. This route reaches the same value and rules out the trap, so the answer is $\{0, 1\}$.

INSIGHT

Trigonometric substitution keeps the square root nonnegative automatically.

№49 Inverse-Sine Domain That Is All Real

••••• L4

FIND THE DOMAIN

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

domain

inverse-sine

range

DOMAIN

$$\mathbb{R}$$

THE TRAP

The trap is $[-1, 1]$, wrongly imposing $|x| \leq 1$ on x instead of on the whole fraction.

Method I: The chapter instrument

1. Use $\left|\frac{2x}{1+x^2}\right| \leq 1$, which is equivalent to $(x-1)^2 \geq 0$ and $(x+1)^2 \geq 0$.
2. Substituting the given restrictions and simplifying gives $\mathbb{R}$.

Method II: Independent check

1. Set $x = \tan(\theta/2)$; the argument becomes $\sin \theta$, always inside $[-1, 1]$.
2. This route reaches the same value and rules out the trap, so the answer is $\mathbb{R}$.

INSIGHT

For inverse functions, the input expression needs the range restriction, not the variable by itself.

№50 Circle Parameter With Missing Point

••••• L4

EVALUATE

Find the point on $X^2 + Y^2 = 1$ represented by $X = \frac{1-t^2}{1+t^2}$, $Y = \frac{2t}{1+t^2}$ when $t = 1$.

parametric

unit-circle

substitution

ANSWER

$$(0, 1)$$

THE TRAP

The trap is $(1, 0)$, plugging $t = 0$ mentally because the formula is familiar.

Method I: The chapter instrument

1. Substitute $t = 1$ directly: $X = (1-1)/(1+1) = 0$ and $Y = 2/(2) = 1$.
2. Substituting the given restrictions and simplifying gives $(0, 1)$.

Method II: Independent check

1. Let $t = \tan(\theta/2)$; $t = 1$ gives $\theta = \pi/2$, hence $(\cos \theta, \sin \theta) = (0, 1)$.
2. This route reaches the same value and rules out the trap, so the answer is $(0, 1)$.

INSIGHT

The rational parameter is slope from $(-1, 0)$, not the angle itself.

№51 Radical Meets Quadratic

••••• L4

SOLVE FOR X

$$\text{Solve for real } x : \quad x + \sqrt{1-x^2} = \sqrt{2}(2x^2 - 1).$$

ANSWER

$$x \in \left\{ \frac{\sqrt{6} + \sqrt{2}}{4}, -\frac{1}{\sqrt{2}} \right\}$$

THE TRAP

Squaring to kill the radical produces a quartic and a swarm of candidate roots; the killer is that squaring forgets $\sqrt{1-x^2} \geq 0$. Indeed the quartic $8x^4 - 4\sqrt{2}x^3 - 6x^2 + 2\sqrt{2}x + 1 = 0$ also offers the phantoms $x = \frac{1}{\sqrt{2}}$ and $x = -\sin \frac{\pi}{12}$, both of which fail the original equation. The matching pitfall in the trig route is writing $\sqrt{1-x^2} = \sin \theta$ for $x = \cos \theta$ — valid only when $\theta \in [0, \pi]$, so $\sin \theta \geq 0$. Students who let θ range freely admit a phantom solution where the radical would be negative.

Cosine substitution with the sign branch

1. Set $x = \cos \theta$ with $\theta \in [0, \pi]$, so $\sqrt{1-x^2} = \sin \theta \geq 0$ and $2x^2 - 1 = \cos 2\theta$; the equation becomes $\cos \theta + \sin \theta = \sqrt{2} \cos 2\theta$.
2. Left side $= \sqrt{2} \sin(\theta + \frac{\pi}{4}) = \sqrt{2} \cos(\frac{\pi}{4} - \theta)$, so $\cos(\frac{\pi}{4} - \theta) = \cos 2\theta$, giving $2\theta = \pm(\frac{\pi}{4} - \theta) + 2k\pi$.
3. The branch $2\theta = \frac{\pi}{4} - \theta + 2k\pi$ gives $\theta = \frac{\pi}{12} + \frac{2k\pi}{3}$, whose values in $[0, \pi]$ are $\theta = \frac{\pi}{12}$ ($k = 0$) and $\theta = \frac{3\pi}{4}$ ($k = 1$); the other branch $2\theta = \theta - \frac{\pi}{4} + 2k\pi$ gives $\theta = -\frac{\pi}{4} + 2k\pi$, none of which lie in $[0, \pi]$. Hence $x = \cos \frac{\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4}$ and $x = \cos \frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$.
4. Both have $\sin \theta \geq 0$ and check in the original equation, so $x \in \left\{ \frac{\sqrt{6}+\sqrt{2}}{4}, -\frac{1}{\sqrt{2}} \right\}$.

Isolate-and-square with validity filter

1. Move the radical alone: $\sqrt{1-x^2} = \sqrt{2}(2x^2 - 1) - x$, which requires the right side ≥ 0 ; squaring yields the quartic $8x^4 - 4\sqrt{2}x^3 - 6x^2 + 2\sqrt{2}x + 1 = 0$.
2. Its four roots are $x = \frac{\sqrt{6}+\sqrt{2}}{4}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\sin \frac{\pi}{12}$; discarding those with $\sqrt{2}(2x^2 - 1) - x < 0$ (the squaring artifacts $\frac{1}{\sqrt{2}}$ and $-\sin \frac{\pi}{12}$) leaves exactly $x = \cos \frac{\pi}{12}$ and $x = -\frac{1}{\sqrt{2}}$.
3. Substituting back confirms both: at $x = -\frac{1}{\sqrt{2}}$, $\text{LHS} = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = 0 = \sqrt{2}(2 \cdot \frac{1}{2} - 1) = \text{RHS}$. Hence $x \in \left\{ \frac{\sqrt{6}+\sqrt{2}}{4}, -\frac{1}{\sqrt{2}} \right\}$.

INSIGHT

For $x = \cos \theta$ the radical $\sqrt{1-x^2}$ is $+\sin \theta$ only on $\theta \in [0, \pi]$; restricting to that window is the trig analogue of keeping the nonnegative square root. Combining $x + \sqrt{1-x^2} = \sqrt{2} \sin(\theta + \frac{\pi}{4})$ with a double-angle right side is a reusable template for radical-plus-quadratic equations.

CHAPTER VI

Concavity, Tangents & Bounds

Concavity, tangent lines, Jensen and SOS bounds.



№52 The Basic Sharp Chain

●●●●● L3

PROVE THAT

$$0 < x < \frac{\pi}{2} \implies \sin x < x < \tan x$$

basic-inequality area tangent

CONCLUSION

$$\sin x < x < \tan x$$

THE TRAP

The trap is assuming the chain is an approximation. It is a strict global inequality on the whole interval.

Method I: The chapter instrument

1. Compare areas in the unit circle sector: inner triangle < sector < tangent triangle.
2. Substituting the given restrictions and simplifying gives $\sin x < x < \tan x$.

Method II: Independent check

1. Study $x - \sin x$ and $\tan x - x$ by derivatives; each starts at 0 and increases on the interval.
2. This route reaches the same value and rules out the trap, so the answer is $\sin x < x < \tan x$.

INSIGHT

This chain is the seed of many sharp trigonometric estimates.

№53 Triangle Sine Sum Ceiling

●●●●● L3

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

Jensen triangle sine

CONCLUSION

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$$

THE TRAP

The trap is 3, maximising each sine independently. The angle sum forces the equilateral ceiling.

Method I: The chapter instrument

1. Use Jensen because $\sin x$ is concave on $(0, \pi)$.
2. Substituting the given restrictions and simplifying gives $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

Method II: Independent check

1. Use $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} \leq 2 \cos \frac{C}{2}$ and optimise the resulting symmetric expression.
2. This route reaches the same value and rules out the trap, so the answer is $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

INSIGHT

Concavity is the official language of 'equal angles are best'.

№54 Sine-Cosine Diagonal

●●●●● L3

FIND THE MAXIMUM

$$\sin x + \cos x \quad (0 \leq x \leq \frac{\pi}{2})$$

maximum

phase-shift

Cauchy

MAXIMUM

$$\sqrt{2}$$

THE TRAP

The trap is 2, trying to make both sine and cosine equal to 1.

Method I: The chapter instrument

1. Write $\sin x + \cos x = \sqrt{2} \sin(x + \pi/4)$.
2. Substituting the given restrictions and simplifying gives $\sqrt{2}$.

Method II: Independent check

1. Apply Cauchy: $(\sin x + \cos x)^2 \leq 2(\sin^2 x + \cos^2 x) = 2$.
2. This route reaches the same value and rules out the trap, so the answer is $\sqrt{2}$.

INSIGHT

The equality point is $x = \pi/4$, where the two components balance.

№55 Acute Tangent Jensen

●●●●● L4

PROVE THAT

$$A, B, C \in (0, \frac{\pi}{2}), A + B + C = \pi \implies \tan A + \tan B + \tan C \geq 3\sqrt{3}$$

Jensen

tangent

acute-triangle

CONCLUSION

$$\tan A + \tan B + \tan C \geq 3\sqrt{3}$$

THE TRAP

The trap is ignoring the word acute; without it tangent can become negative and Jensen on one convex interval is no longer valid.

Method I: The chapter instrument

1. On $(0, \pi/2)$, $\tan x$ is convex, so Jensen gives the minimum at $A = B = C = \pi/3$.
2. Substituting the given restrictions and simplifying gives $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.

Method II: Independent check

1. Use $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ and AM-GM on the positive tangents.
2. This route reaches the same value and rules out the trap, so the answer is $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.

INSIGHT

The identity and the inequality cooperate only when all tangents are positive.

№56 Cosine Sum Triangle Bound

••••• L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \cos A + \cos B + \cos C \leq \frac{3}{2}$$

triangle cosine sharp-bound

CONCLUSION

$$\cos A + \cos B + \cos C \leq \frac{3}{2}$$

THE TRAP

The tempting bound 3 is true but not sharp. The ranker task is finding the equality case.

Method I: The chapter instrument

1. Use $\cos A + \cos B + \cos C = 1 + r/R$ and Euler's $r \leq R/2$.
2. Substituting the given restrictions and simplifying gives $\cos A + \cos B + \cos C \leq \frac{3}{2}$.

Method II: Independent check

1. Use product-to-sum to reduce to $1 + 4 \sin(A/2) \sin(B/2) \sin(C/2)$, then maximise the product.
2. This route reaches the same value and rules out the trap, so the answer is $\cos A + \cos B + \cos C \leq \frac{3}{2}$.

INSIGHT

Many triangle inequalities hide a radius inequality.

№57 Product of Sines Ceiling

••••• L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$$

product Jensen triangle

CONCLUSION

$$\sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$$

THE TRAP

The trap is 1, attainable by one sine at a time but never by all three under the sum constraint.

Method I: The chapter instrument

1. Apply Jensen to $\log \sin x$ on $(0, \pi)$.
2. Substituting the given restrictions and simplifying gives $\sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$.

Method II: Independent check

1. Use smoothing: replacing two unequal angles by their average increases the product while preserving the sum.
2. This route reaches the same value and rules out the trap, so the answer is $\sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$.

INSIGHT

Products under a fixed sum usually ask for logarithmic concavity.

№58 Cosine-Square Floor

••••• L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \cos^2 A + \cos^2 B + \cos^2 C \geq \frac{3}{4}$$

cosine-square

triangle

minimum

CONCLUSION

$$\cos^2 A + \cos^2 B + \cos^2 C \geq \frac{3}{4}$$

THE TRAP

The trap is 0, imagining all cosines can vanish. Three positive angles summing to π cannot all be right angles.

Method I: The chapter instrument

1. Use $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2 \cos A \cos B \cos C$ and maximise the product of cosines in the acute case.
2. Substituting the given restrictions and simplifying gives $\cos^2 A + \cos^2 B + \cos^2 C \geq \frac{3}{4}$.

Method II: Independent check

1. By uvw/smoothing, the minimum occurs at $A = B = C = \pi/3$, giving $3 \cdot (1/2)^2$.
2. This route reaches the same value and rules out the trap, so the answer is $\cos^2 A + \cos^2 B + \cos^2 C \geq \frac{3}{4}$.

INSIGHT

The obtuse case only makes the product of cosines nonpositive, so it cannot lower the sum.

№59 Sine-Square Ceiling

●●●●● L4

PROVE THAT

$$A, B, C > 0, A + B + C = \pi \implies \sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$$

sine-square

triangle

maximum

CONCLUSION

$$\sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$$

THE TRAP

The trap is 3, the separate bound for three unrelated sines.

Method I: The chapter instrument

1. Use $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C$ and maximise the cosine product.
2. Substituting the given restrictions and simplifying gives $\sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$.

Method II: Independent check

1. Apply Jensen to $\log \sin x$ indirectly, or smooth two angles to show the maximum is equilateral.
2. This route reaches the same value and rules out the trap, so the answer is $\sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$.

INSIGHT

The square-sum maximum and product maximum occur at the same balanced triangle.

№60 Two-Term AM-GM in Disguise

●●●●● L3

PROVE THAT

$$0 < x < \frac{\pi}{2} \implies \tan x + \cot x \geq 2$$

AM-GM

tangent

reciprocal

CONCLUSION

$$\tan x + \cot x \geq 2$$

THE TRAP

The trap is 1, from thinking reciprocal terms cancel. Positive reciprocals add, they do not cancel.

Method I: The chapter instrument

1. Let $t = \tan x > 0$; then the expression is $t + 1/t \geq 2$.
2. Substituting the given restrictions and simplifying gives $\tan x + \cot x \geq 2$.

Method II: Independent check

1. Use $\tan x + \cot x = 1/(\sin x \cos x)$ and $\sin x \cos x \leq 1/2$.
2. This route reaches the same value and rules out the trap, so the answer is $\tan x + \cot x \geq 2$.

INSIGHT

A reciprocal trig expression often becomes one-line AM-GM after the right substitution.

№61 Endpoint Tangent Line

●●●●● L5

PROVE THAT

$$0 \leq x \leq \frac{\pi}{2} \implies \sin x \geq \frac{2x}{\pi}$$

concavity

chord

sharp-bound

CONCLUSION

$$\sin x \geq \frac{2x}{\pi}$$

THE TRAP

The trap is reversing the inequality because $\sin x \leq x$. A chord bound and a tangent bound point in opposite directions.

Method I: The chapter instrument

1. Since $\sin x$ is concave on $[0, \pi]$, it lies above the chord joining $(0, 0)$ and $(\pi/2, 1)$.
2. Substituting the given restrictions and simplifying gives $\sin x \geq \frac{2x}{\pi}$.

Method II: Independent check

1. Consider $f(x) = \sin x - 2x/\pi$; its minimum on the interval occurs at an endpoint after checking the single critical point.
2. This route reaches the same value and rules out the trap, so the answer is $\sin x \geq \frac{2x}{\pi}$.

INSIGHT

Concavity gives lower chord bounds and upper tangent bounds; mixing them is a classic trap.

CHAPTER VII

Phase, Range & Extremum Machines

R

Phase shifts and envelopes find the range before calculus.

№62 Three-Four-Five Phase

●●●●● L3

FIND THE RANGE

$$3 \sin x + 4 \cos x$$

phase-shift

range

R-method

RANGE

$$[-5, 5]$$

THE TRAP

The trap is $[-7, 7]$, adding absolute coefficients. The two components cannot peak together unless their phases match.

Method I: The chapter instrument

1. Write $3 \sin x + 4 \cos x = 5 \sin(x + \phi)$ where $\cos \phi = 3/5$ and $\sin \phi = 4/5$.
2. Substituting the given restrictions and simplifying gives $[-5, 5]$.

Method II: Independent check

1. Apply Cauchy: $|3 \sin x + 4 \cos x| \leq 5 \sqrt{\sin^2 x + \cos^2 x} = 5$, and equality is attainable.
2. This route reaches the same value and rules out the trap, so the answer is $[-5, 5]$.

INSIGHT

The R-method is just Cauchy with equality encoded as a phase.

№63 Shifted Thirteen

●●●●● L3

FIND THE RANGE

$$2 + 5 \sin x - 12 \cos x$$

range

phase-shift

constant

RANGE

$$[-11, 15]$$

THE TRAP

The trap is $[-19, 19]$, forgetting the constant shift after finding amplitude 13.

Method I: The chapter instrument

1. The oscillating part has amplitude $\sqrt{5^2 + (-12)^2} = 13$, then shift by 2.
2. Substituting the given restrictions and simplifying gives $[-11, 15]$.

Method II: Independent check

1. Square the distance from $(\sin x, \cos x)$ to the coefficient vector and use Cauchy.
2. This route reaches the same value and rules out the trap, so the answer is $[-11, 15]$.

INSIGHT

Always add the vertical shift after finding amplitude.

№64 Quadratic Form on the Circle

●●●●● L4

FIND THE RANGE

$$\sin^2 x + 2 \sin x \cos x + 3 \cos^2 x$$

quadratic-form

eigenvalue

range

RANGE

$$[2 - \sqrt{2}, 2 + \sqrt{2}]$$

THE TRAP

The trap is $[1, 3]$, reading only the diagonal coefficients and ignoring the cross term.

Method I: The chapter instrument

1. Write the expression as $2 - \cos 2x + \sin 2x$, whose oscillating part has amplitude $\sqrt{2}$.
2. Substituting the given restrictions and simplifying gives $[2 - \sqrt{2}, 2 + \sqrt{2}]$.

Method II: Independent check

1. View it as the quadratic form of the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$ on the unit circle; its eigenvalues are $2 \pm \sqrt{2}$.
2. This route reaches the same value and rules out the trap, so the answer is $[2 - \sqrt{2}, 2 + \sqrt{2}]$.

INSIGHT

Cross terms rotate the axes; eigenvalues are the true extrema.

№65 Square-Difference Wave

●●●●● L4

FIND THE RANGE

$$4 \sin x \cos x + 3(\cos^2 x - \sin^2 x)$$

double-angle

range

phase

RANGE

$$[-\sqrt{13}, \sqrt{13}]$$

THE TRAP

The trap is $[-7, 7]$, adding amplitudes before recognising a single $2x$ -wave.

Method I: The chapter instrument

1. Convert to $2 \sin 2x + 3 \cos 2x$, whose amplitude is $\sqrt{13}$.
2. Substituting the given restrictions and simplifying gives $[-\sqrt{13}, \sqrt{13}]$.

Method II: Independent check

1. Use Cauchy on the vector $(\sin 2x, \cos 2x)$ with coefficients $(2, 3)$.
2. This route reaches the same value and rules out the trap, so the answer is $[-\sqrt{13}, \sqrt{13}]$.

INSIGHT

№66 Nonnegative Shift

●●●●● L3

FIND THE RANGE

$$5 + 3 \sin x + 4 \cos x$$

range phase-shift amplitude

RANGE

$$[0, 10]$$

THE TRAP

The trap is $[-5, 5]$, finding only the range of the oscillating part and forgetting the $+5$.

Method I: The chapter instrument

1. The oscillating part has amplitude 5, so adding 5 gives $[0, 10]$.
2. Substituting the given restrictions and simplifying gives $[0, 10]$.

Method II: Independent check

1. Complete the phase: $3 \sin x + 4 \cos x = 5 \sin(x + \phi)$.
2. This route reaches the same value and rules out the trap, so the answer is $[0, 10]$.

INSIGHT

A nonnegative-looking expression may simply be a shifted sinusoid.

№67 A Sinusoidal Equation Parameter

●●●●● L4

EVALUATE

Find all real k for which $\sin x + \cos x = k$ has a real solution.

parameter range phase-shift

ANSWER

$$[-\sqrt{2}, \sqrt{2}]$$

THE TRAP

The trap is $[-2, 2]$, adding the separate ranges of sine and cosine.

Method I: The chapter instrument

1. Write $\sin x + \cos x = \sqrt{2} \sin(x + \pi/4)$.
2. Substituting the given restrictions and simplifying gives $[-\sqrt{2}, \sqrt{2}]$.

Method II: Independent check

1. Square after rearranging: $(\sin x + \cos x)^2 = 1 + \sin 2x$ ranges from 0 to 2, with both signs attainable.
2. This route reaches the same value and rules out the trap, so the answer is $[-\sqrt{2}, \sqrt{2}]$.

INSIGHT

Parameter existence questions are range questions in disguise.

№68 Sine Product Peak

●●●●● L3

FIND THE MAXIMUM

$$\sin x \cos x$$

maximum double-angle product

MAXIMUM

$$\frac{1}{2}$$

THE TRAP

The trap is 1, trying to maximise both factors independently.

Method I: The chapter instrument

1. Use $\sin x \cos x = \frac{1}{2} \sin 2x$.
2. Substituting the given restrictions and simplifying gives $\frac{1}{2}$.

Method II: Independent check

1. Use $2ab \leq a^2 + b^2$ with $a = \sin x, b = \cos x$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{1}{2}$.

INSIGHT

Products of two phase-shifted unit components peak when the components balance.

№69 Secant-Tangent Square Range

••••• L4

FIND THE RANGE

$$\frac{1 + \sin x}{1 - \sin x} \quad (x \neq \frac{\pi}{2} + 2k\pi)$$

range half-angle unbounded

RANGE

$$[0, \infty)$$

THE TRAP

The trap is $[-1, 1]$, treating the quotient like a sine. The denominator can approach zero.

Method I: The chapter instrument

1. Rewrite as $\left(\frac{1 + \sin x}{\cos x}\right)^2 = (\sec x + \tan x)^2$ where defined.
2. Substituting the given restrictions and simplifying gives $[0, \infty)$.

Method II: Independent check

1. Let $t = \tan(x/2)$; the expression becomes $\left(\frac{1+t}{1-t}\right)^2$, which covers all nonnegative values.
2. This route reaches the same value and rules out the trap, so the answer is $[0, \infty)$.

INSIGHT

A quotient of shifted sine terms often becomes a perfect square.

№70 Quadratic Sine Range

••••• L4

FIND THE RANGE

$$2 \sin^2 x - 2 \sin x + 3$$

quadratic range vertex

RANGE

$$\left[\frac{5}{2}, 7\right]$$

THE TRAP

The trap is $[3, 7]$, checking only endpoints of $\sin x$ and missing the vertex inside.

Method I: The chapter instrument

1. Let $u = \sin x \in [-1, 1]$; then the expression is $2u^2 - 2u + 3 = 2(u - 1/2)^2 + 5/2$.
2. Substituting the given restrictions and simplifying gives $[\frac{5}{2}, 7]$.

Method II: Independent check

1. Differentiate with respect to x and check both critical points and endpoints.
2. This route reaches the same value and rules out the trap, so the answer is $[\frac{5}{2}, 7]$.

INSIGHT

After substituting $u = \sin x$, remember the restricted interval for u .

71 Two-Variable Symmetric Peak

••••• L5

FIND THE MAXIMUM

$$\sin x + \sin y \quad \text{given } x + y = \frac{2\pi}{3}$$

two-variable

sum-to-product

maximum

MAXIMUM

$$\sqrt{3}$$

THE TRAP

The trap is 2, maximising the two sines independently without respecting the sum.

Method I: The chapter instrument

1. Use $\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} = \sqrt{3} \cos \frac{x-y}{2}$.
2. Substituting the given restrictions and simplifying gives $\sqrt{3}$.

Method II: Independent check

1. By Jensen, sine is concave on the relevant balanced interval, so the maximum occurs at $x = y = \pi/3$.
2. This route reaches the same value and rules out the trap, so the answer is $\sqrt{3}$.

INSIGHT

Sum-to-product converts the constraint into a single free difference.

CHAPTER VIII

Periodicity & Root-Counting

Periodicity and intervals separate true roots from impostors.

№72 Double Sine Roots

●●●●● L3

SOLVE

$$\sin 2x = \sin x \quad (0 \leq x < 2\pi)$$

equation

root-counting

double-angle

SOLUTIONS

$$\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$$

THE TRAP

The trap is dividing by $\sin x$ and losing $x = 0, \pi$.

Method I: The chapter instrument

1. Use $\sin 2x = 2 \sin x \cos x$, then factor $\sin x(2 \cos x - 1) = 0$.
2. Substituting the given restrictions and simplifying gives $\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.

Method II: Independent check

1. Use $\sin u = \sin v$ with $u = 2x, v = x$ and merge the two resulting arithmetic progressions.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.

INSIGHT

Never divide by a trigonometric factor before recording its zeros.

№73 Cosine Double Roots

●●●●● L3

SOLVE

$$\cos 2x = \cos x \quad (0 \leq x < 2\pi)$$

equation

cosine

quadratic

SOLUTIONS

$$\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$$

THE TRAP

The trap is $x = 0$ only, caused by cancelling cosine too early.

Method I: The chapter instrument

1. Set $c = \cos x$ and solve $2c^2 - 1 = c$.
2. Substituting the given restrictions and simplifying gives $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

Method II: Independent check

1. Use $\cos u = \cos v$, so $u = \pm v + 2k\pi$, and restrict to the interval.
2. This route reaches the same value and rules out the trap, so the answer is $\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\}$.

INSIGHT

Cosine equations often become polynomial equations in $\cos x$.

№74 Triple Zero Count

●●●●● L3

SOLVE

$$\sin 3x = 0 \quad (0 \leq x < 2\pi)$$

periodicity

zero-counting

sine

SOLUTIONS

$$\left\{ \frac{k\pi}{3} : 0 \leq k \leq 5 \right\}$$

THE TRAP

The trap is listing only $0, \pi$, the zeros of $\sin x$ rather than $\sin 3x$.

Method I: The chapter instrument

1. Set $3x = k\pi$ and impose $0 \leq x < 2\pi$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{k\pi}{3} : 0 \leq k \leq 5 \right\}$.

Method II: Independent check

1. Observe that $\sin 3x$ completes three full sine periods on $[0, 2\pi)$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{k\pi}{3} : 0 \leq k \leq 5 \right\}$.

INSIGHT

Counting periods before solving prevents missing repeated waves.

№75 Quadratic Cosine Equation

●●●●● L3

SOLVE

$$2 \cos^2 x - 3 \cos x + 1 = 0 \quad (0 \leq x < 2\pi)$$

quadratic

cosine

interval

SOLUTIONS

$$\left\{ 0, \frac{\pi}{3}, \frac{5\pi}{3} \right\}$$

THE TRAP

The trap is keeping only the principal angle $\pi/3$ and forgetting the fourth-quadrant cosine.

Method I: The chapter instrument

1. Factor $(2 \cos x - 1)(\cos x - 1) = 0$ and solve each cosine equation on the interval.
2. Substituting the given restrictions and simplifying gives $\left\{ 0, \frac{\pi}{3}, \frac{5\pi}{3} \right\}$.

Method II: Independent check

1. Graph $y = \cos x$ over one full period and read off where it takes values 1 and $1/2$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ 0, \frac{\pi}{3}, \frac{5\pi}{3} \right\}$.

INSIGHT

After solving for a trig value, the interval still controls how many angles appear.

№76 Sine Five Meets Sine Three

●●●●● L4

COUNT

Number of solutions of $\sin 5x = \sin 3x$ on $[0, 2\pi)$.

counting

product-to-sum

frequency

COUNT

10

THE TRAP

The trap is 2, from cancelling the common-looking sine. These are different frequencies.

Method I: The chapter instrument

1. Use $\sin 5x - \sin 3x = 2 \cos 4x \sin x$; count zeros of both factors without overlap.
2. Substituting the given restrictions and simplifying gives 10.

Method II: Independent check

1. Use $\sin u = \sin v$: $5x = 3x + 2k\pi$ or $5x = \pi - 3x + 2k\pi$, then count interval-valid roots.
2. This route reaches the same value and rules out the trap, so the answer is 10.

INSIGHT

Product-to-sum turns frequency equations into clean zero counts.

№77 Cosine Plus Cosine Two

●●●●● L4

SOLVE

$$\cos x + \cos 2x = 0 \quad (0 \leq x < 2\pi)$$

equation

cosine

quadratic

SOLUTIONS

$$\left\{ \frac{\pi}{3}, \pi, \frac{5\pi}{3} \right\}$$

THE TRAP

The trap is solving $\cos x = 0$ and $\cos 2x = 0$ separately; a sum is not a product.

Method I: The chapter instrument

1. Let $c = \cos x$; then $c + 2c^2 - 1 = 0$, so $c = 1/2$ or $c = -1$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{\pi}{3}, \pi, \frac{5\pi}{3} \right\}$.

Method II: Independent check

1. Use product-to-sum: $\cos x + \cos 2x = 2 \cos(3x/2) \cos(x/2)$ and solve both factors.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{\pi}{3}, \pi, \frac{5\pi}{3} \right\}$.

INSIGHT

A sum can be made into a product before root-counting.

№78 Sine Equals Cosine Double

●●●●● L4

SOLVE

$$\sin x = \cos 2x \quad (0 \leq x < 2\pi)$$

equation sine double-angle

SOLUTIONS

$$\left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \right\}$$

THE TRAP

The trap is to square both sides and introduce extra roots. No squaring is needed.

Method I: The chapter instrument

1. Write $\cos 2x = 1 - 2\sin^2 x$ and solve $2s^2 + s - 1 = 0$ for $s = \sin x$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \right\}$.

Method II: Independent check

1. Graph $y = \sin x$ and $y = \cos 2x$ over one period to verify the three intersections.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \right\}$.

INSIGHT

Choosing the variable already present in both sides avoids extraneous roots.

№79 Tangent Reciprocal Equation

●●●●● L4

SOLVE

$$\tan x + \cot x = 4 \quad \left(0 < x < \frac{\pi}{2}\right)$$

tangent reciprocal quadratic

SOLUTIONS

$$\left\{ \frac{\pi}{12}, \frac{5\pi}{12} \right\}$$

THE TRAP

The trap is $x = \pi/4$, where the left side is 2, not 4.

Method I: The chapter instrument

1. Let $t = \tan x > 0$; solve $t + 1/t = 4$, giving $t = 2 \pm \sqrt{3}$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{\pi}{12}, \frac{5\pi}{12} \right\}$.

Method II: Independent check

1. Use $\tan(\pi/12) = 2 - \sqrt{3}$ and $\tan(5\pi/12) = 2 + \sqrt{3}$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{\pi}{12}, \frac{5\pi}{12} \right\}$.

INSIGHT

Reciprocal tangent equations usually come in complementary pairs.

№80 Zeros on a Closed Half-Period

●●●●● L3

COUNT

For positive integer n , count the roots of $\sin nx = 0$ on $[0, \pi]$.

counting endpoint periodicity

COUNT

$$n + 1$$

THE TRAP

The trap is n , forgetting that a closed interval includes both endpoints.

Method I: The chapter instrument

1. Solve $nx = k\pi$; the interval gives $0 \leq k \leq n$.
2. Substituting the given restrictions and simplifying gives $n + 1$.

Method II: Independent check

1. View $\sin nx$ as n half-waves on $[0, \pi]$, with one more node than half-waves.
2. This route reaches the same value and rules out the trap, so the answer is $n + 1$.

INSIGHT

Endpoint conventions are a serious source of off-by-one errors.

№81 Two-Equation Unit Sum

••••• L5

SOLVE

$$\sin x + \sin y = 1, \quad \cos x + \cos y = 1 \quad (0 \leq x, y < 2\pi)$$

system

sum-to-product

two-variable

SOLUTIONS

$$\left\{ \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, 0\right) \right\}$$

THE TRAP

The trap is $(\pi/4, \pi/4)$, because each equation would then equal $\sqrt{2}$, not 1.

Method I: The chapter instrument

1. Use sum-to-product with $S = (x + y)/2$, $D = (x - y)/2$: $2 \sin S \cos D = 1$ and $2 \cos S \cos D = 1$.
2. Substituting the given restrictions and simplifying gives $\left\{ \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, 0\right) \right\}$.

Method II: Independent check

1. Subtract the equations after converting to vectors: two unit vectors sum to $(1, 1)$, forcing the vectors $(1, 0)$ and $(0, 1)$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, 0\right) \right\}$.

INSIGHT

Vector geometry is often the shortest way to solve paired sine-cosine systems.

№82 Where the Line Meets the Wave

••••• L4

FIND THE NUMBER OF SOLUTIONS

Find the total number of real solutions of $\sin x = \frac{x}{10\pi}$.

bounding

graphical-count

line-vs-sine

symmetry

ANSWER

19 real solutions.

THE TRAP

The fatal shortcut is 'each period gives two intersections, and 10π on each side spans 5 periods, so $2 \cdot 5 \cdot 2 = 20$.' This over-counts the very first hump. For $x > 0$ the line $y = \frac{x}{10\pi}$ is positive, so it can only meet the sine on the five upward humps $(2k\pi, (2k+1)\pi)$; the downward dips contribute nothing. Four of those upward humps are cut twice, but the first hump $(0, \pi)$ shares the origin with the line and is cut only once (sine leaves 0 with slope $1 > \frac{1}{10\pi}$, overshoots, and returns once). So each side gives $4 \cdot 2 + 1 = 9$, not 10. Careful bounding (the line exits $[-1, 1]$ exactly at $x = \pm 10\pi$) gives 19, not 20.

Bound the window, then count humps

1. Since $|\sin x| \leq 1$, any solution needs $|\frac{x}{10\pi}| \leq 1$, i.e. $x \in [-10\pi, 10\pi]$.
2. The function is odd (both sides change sign under $x \mapsto -x$), and $x = 0$ is a solution, so count solutions in $(0, 10\pi]$ and double.
3. On $(0, 10\pi)$ the line $y = \frac{x}{10\pi}$ is positive, so it can only cross the sine where $\sin x > 0$: the five upward humps on $(2k\pi, (2k+1)\pi)$ for $k = 0, 1, 2, 3, 4$. The four humps with $k \geq 1$ each rise to 1 while the line stays ≤ 0.9 , so each is cut twice (going up and coming down). The first hump $(0, \pi)$ shares the origin with the line: sine departs 0 with slope $1 > \frac{1}{10\pi}$, climbs above the line, and meets it again exactly once, so it gives one crossing in $(0, \pi)$. The down-dips on $((2k+1)\pi, (2k+2)\pi)$ have $\sin x < 0 < \frac{x}{10\pi}$ and contribute nothing.
4. Hence positive roots $= 4 \cdot 2 + 1 = 9$; by oddness 9 negative; plus the origin: $9 + 9 + 1 = \boxed{19}$.

Sign-change accounting

1. Let $g(x) = \sin x - \frac{x}{10\pi}$; roots of g are the solutions. Outside $[-10\pi, 10\pi]$, g keeps the sign of $-\frac{x}{10\pi}$ and never returns to 0.
2. For $x > 0$, $g < 0$ on every down-dip $((2k+1)\pi, (2k+2)\pi)$ (both terms negative), so all positive roots sit on the upward humps. On each upward hump g runs $- \rightarrow + \rightarrow -$, giving two sign changes — except the first hump, where $g(0^+) = \sin x - \frac{x}{10\pi} > 0$ already (slope 1 beats slope $\frac{1}{10\pi}$), so it shows only one sign change before $g(\pi^-) = -\frac{1}{10} < 0$.
3. Counting sign changes on $(0, 10\pi]$ gives $2 \cdot 4 + 1 = 9$; by oddness the count on $[-10\pi, 0)$ is also 9; the simple root at $x = 0$ adds one, for $\boxed{19}$.

INSIGHT

Replacing 10π by $2N\pi$ turns this into 'how many times does the line $y = \frac{x}{2N\pi}$ cross $\sin x$?' With the cutoff an exact even multiple of π , the line just touches ± 1 at the endpoints, the down-dips never qualify, N upward humps per side give $2N$ crossings — but the origin-sharing first hump loses one — so each side contributes $2N - 1$ and the total is the exact odd integer $2(2N - 1) + 1 = 4N - 1$. Here $N = 5$ gives 19; that single missing crossing at the first hump is exactly what defeats the naive ' $4N$ '.

№83 Three Tangents in a Row

••••• L4

SOLVE FOR X

Solve $\tan x + \tan 2x + \tan 3x = 0$ for $x \in (0, \pi)$, stating how many solutions there are.

tangent

factoring

chebyshev

domain-restriction

ANSWER

$$x = \frac{\pi}{3}, \frac{2\pi}{3} \text{ or } \cos^2 x = \frac{2}{3} \left(\cos x = \pm \frac{\sqrt{6}}{3} \right); \quad \boxed{4} \text{ solutions in } (0, \pi).$$

THE TRAP

The famous identity $\tan x + \tan 2x + \tan 3x = \tan x \tan 2x \tan 3x$ is false here — it holds only when the angles sum to a multiple of π , and $x + 2x + 3x = 6x$ is not. A solver who blindly applies it (or who naively writes $\tan 3x = \tan(x + 2x)$ and cross-multiplies, dropping the points where $\cos x, \cos 2x, \cos 3x$ vanish) corrupts the root set. The correct route keeps the full domain restriction $x \notin \{\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{6}\}$.

Clear denominators into a single fraction

1. Combine over a common denominator: $\tan x + \tan 2x + \tan 3x = \frac{\sin 3x (3 \cos 3x + \cos x)}{2 \cos x \cos 2x \cos 3x}$ (verified by expanding $\tan x + \tan 2x = \frac{\sin 3x}{\cos x \cos 2x}$ and adding $\tan 3x$).
2. Numerator zero requires $\sin 3x = 0$ or $3 \cos 3x + \cos x = 0$. Using $\cos 3x = 4 \cos^3 x - 3 \cos x$, the second factor is $12 \cos^3 x - 8 \cos x = 4 \cos x (3 \cos^2 x - 2)$.
3. Since $\cos x = 0$ is forbidden (it kills $\tan x$), the second family is $3 \cos^2 x = 2$, i.e. $\cos x = \pm \frac{\sqrt{6}}{3}$ — two roots in $(0, \pi)$.
4. From $\sin 3x = 0$: $x = \frac{k\pi}{3}$, giving $x = \frac{\pi}{3}, \frac{2\pi}{3}$ in $(0, \pi)$ (both lie in the domain). Total $2 + 2 = \boxed{4}$.

Chebyshev in $\cos x$

1. Multiply through by $\cos x \cos 2x \cos 3x$ (nonzero on the domain) to reach $\sin 3x (3 \cos 3x + \cos x) = 0$.
2. Write everything in $c = \cos x$: $\sin 3x = \sin x (3 - 4 \sin^2 x) = \sin x (4c^2 - 1)$ and $3 \cos 3x + \cos x = 12c^3 - 8c = 4c(3c^2 - 2)$.
3. So the product vanishes when $\sin x = 0$ (excluded as endpoints), $4c^2 - 1 = 0$ ($c = \pm \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{2\pi}{3}$), or $3c^2 - 2 = 0$ ($c = \pm \frac{\sqrt{6}}{3}$).
4. Collecting the valid roots in $(0, \pi)$: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \arccos \frac{\sqrt{6}}{3}, \pi - \arccos \frac{\sqrt{6}}{3}$ — exactly $\boxed{4}$.

INSIGHT

The clean numerator $\sin 3x (3 \cos 3x + \cos x)$ generalises: for $\tan x + \tan 2x + \dots + \tan nx$ one always lands on $\sin(\text{top})$ times a Chebyshev polynomial in $\cos x$, so the solution set splits into 'evenly spaced' zeros and algebraic roots — and the domain exclusions (where any tan blows up) must be subtracted by hand.

№84 The Reachable Heights of k

●●●●● L4

FIND THE RANGE

Find all real k for which $\sin x + \cos x + \sin x \cos x = k$ has a real solution; and say how many solutions $x \in [0, 2\pi)$ occur when k is least.

parameter substitution range solvability

ANSWER

Solvable $\iff k \in \left[-1, \frac{1+2\sqrt{2}}{2}\right]$; at the minimum $k = -1$ there are $\boxed{2}$ solutions in $[0, 2\pi)$.

THE TRAP

After substituting $t = \sin x + \cos x$ and $\sin x \cos x = \frac{t^2-1}{2}$, the equation becomes $k = \frac{t^2+2t-1}{2}$ — but the trap is taking t over all of $\mathbb{R}$. In fact $t = \sqrt{2} \sin(x + \frac{\pi}{4})$ is confined to $[-\sqrt{2}, \sqrt{2}]$, and the vertex of the parabola sits at $t = -1$ (interior), so the minimum of k is -1 — not the parabola's value at an endpoint. Forgetting the constraint $|t| \leq \sqrt{2}$ produces a bogus, unbounded range.

Substitute and optimise on the true interval

1. Let $t = \sin x + \cos x$, so $\sin x \cos x = \frac{t^2-1}{2}$ and $t \in [-\sqrt{2}, \sqrt{2}]$.
2. Then $k = t + \frac{t^2-1}{2} = \frac{1}{2}(t^2 + 2t - 1) = \frac{1}{2}((t+1)^2 - 2)$.
3. On $[-\sqrt{2}, \sqrt{2}]$ this parabola has minimum at $t = -1$: $k = \frac{1}{2}(0 - 2) = -1$, and maximum at the right endpoint $t = \sqrt{2}$: $k = \frac{1}{2}(2 + 2\sqrt{2} - 1) = \frac{1+2\sqrt{2}}{2}$.
4. So k ranges over $\left[-1, \frac{1+2\sqrt{2}}{2}\right]$. At $k = -1$ we need $t = -1$, i.e. $\sqrt{2} \sin(x + \frac{\pi}{4}) = -1 \Rightarrow \sin(x + \frac{\pi}{4}) = -\frac{1}{\sqrt{2}}$, giving $x = \pi, \frac{3\pi}{2}$ — $\boxed{2}$ solutions.

Solvability as a quadratic-in- t feasibility

1. The relation $t^2 + 2t - (1 + 2k) = 0$ must have a root $t \in [-\sqrt{2}, \sqrt{2}]$ for the equation to be solvable.
2. Solving, $t = -1 \pm \sqrt{2+2k}$ (real iff $k \geq -1$). The branch $t = -1 + \sqrt{2+2k}$ lands in $[-\sqrt{2}, \sqrt{2}]$ precisely when $\sqrt{2+2k} \leq 1 + \sqrt{2}$, i.e. $k \leq \frac{(1+\sqrt{2})^2-2}{2} = \frac{1+2\sqrt{2}}{2}$.
3. Hence the feasibility window is $k \in \left[-1, \frac{1+2\sqrt{2}}{2}\right]$.
4. At the lower end $k = -1$, the only admissible t is $t = -1$ (a double root), and $\sin x + \cos x = -1$ has exactly $x = \pi, \frac{3\pi}{2}$ in $[0, 2\pi)$ — so $\boxed{2}$ solutions.

INSIGHT

The whole problem is the geometry of $k = \frac{1}{2}((t+1)^2 - 2)$ restricted to a chord $|t| \leq \sqrt{2}$: because the vertex $t = -1$ is interior, the minimum is attained inside and is 'clean', while the maximum is forced to an endpoint and carries the surd $\sqrt{2}$. Any 'sum + product' trig equation collapses to such a parabola in $t = \sin x \pm \cos x$.

№85 When the Sines Equal the Cosines

●●●●● L4

SOLVE FOR x

Solve $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$ for $x \in [0, 2\pi)$, and count the solutions.

sum-to-product factoring mixed-equation solution-counting

ANSWER

$$x = 2k\pi \pm \frac{2\pi}{3} \text{ or } x = \frac{\pi}{8} + \frac{k\pi}{2}; \quad \boxed{6} \text{ solutions in } [0, 2\pi).$$

THE TRAP

The instinct is to move everything to one side and grind through six sum-to-product steps, where a single sign slip wrecks the count. The elegant factorisations of each side share the factor $(2 \cos x + 1)$, but a careless solver cancels $(2 \cos x + 1)$ to get only $\tan 2x = 1$, thereby discarding the two genuine roots from $\cos x = -\frac{1}{2}$. Cancelling a factor that can legitimately vanish is the classic root-loss.

Factor each side, then divide cases

1. Use the trinity factorisations: LHS = $\sin 2x(2 \cos x + 1)$, RHS = $\cos 2x(2 \cos x + 1)$.
2. Subtract: $(2 \cos x + 1)(\sin 2x - \cos 2x) = 0$, so $2 \cos x + 1 = 0$ or $\sin 2x = \cos 2x$.
3. $2 \cos x + 1 = 0 \Rightarrow \cos x = -\frac{1}{2} \Rightarrow x = \frac{2\pi}{3}, \frac{4\pi}{3}$ — two roots.
4. $\sin 2x = \cos 2x \Rightarrow \tan 2x = 1 \Rightarrow 2x = \frac{\pi}{4} + k\pi \Rightarrow x = \frac{\pi}{8} + \frac{k\pi}{2}$, giving $x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$ — four roots. Total $\boxed{6}$.

Single-fraction Dirichlet form

1. Both sides are Dirichlet sums: $\sum_{j=1}^3 \sin jx = \sin 2x(1 + 2 \cos x)$ and $\sum_{j=1}^3 \cos jx = \cos 2x(1 + 2 \cos x)$ (the common factor $1 + 2 \cos x = \frac{\sin(3x/2)}{\sin(x/2)}$).
2. The equation becomes $(1 + 2 \cos x)(\sin 2x - \cos 2x) = 0$; writing $\sin 2x - \cos 2x = \sqrt{2} \sin(2x - \frac{\pi}{4})$, the second factor vanishes when $2x - \frac{\pi}{4} = k\pi$.
3. Solve: $x = \frac{\pi}{8} + \frac{k\pi}{2}$ (four values in $[0, 2\pi)$) together with $\cos x = -\frac{1}{2}$ (two values).
4. Grand total $4 + 2 = \boxed{6}$ solutions.

INSIGHT

Subtracting two Dirichlet kernels of the same length always leaves their shared factor $\frac{\sin(nx/2)}{\sin(x/2)}$ multiplied by a difference of the leading harmonics. Recognising the shared factor turns a twelve-term grind into a two-case split — and the discipline of keeping $\cos x = -\frac{1}{2}$ guards against the cancellation trap.

№86 A Product Pinned to One

●●●●● L5

FIND THE NUMBER OF SOLUTIONS

Find the number of solutions in $[0, 2\pi]$ of $\cos x \cos 2x \cos 3x = 1$.

product

extremal

bounding

solution-counting

ANSWER

$$x = k\pi, \text{ so } x \in \{0, \pi, 2\pi\}; \quad \boxed{3} \text{ solutions in } [0, 2\pi].$$

THE TRAP

The seductive move is to expand $\cos x \cos 2x \cos 3x$ into a sum of cosines and chase the resulting polynomial in $\cos x$ — a brutal degree-6 grind that invites algebra errors. The structural insight is that a product of three quantities each in $[-1, 1]$ can equal $+1$ only if each factor is ± 1 with an even number of -1 's. Skipping this and 'solving' the expanded equation numerically risks missing that the constraints $|\cos x| = |\cos 2x| = |\cos 3x| = 1$ must hold simultaneously.

Extremal pinning

1. Each of $\cos x, \cos 2x, \cos 3x$ lies in $[-1, 1]$, so $|\cos x \cos 2x \cos 3x| \leq 1$ with equality only if all three have absolute value 1.
2. $|\cos x| = 1 \Rightarrow x = k\pi$; check that at $x = k\pi$ also $|\cos 2x| = |\cos 3x| = 1$ automatically (since $2k\pi, 3k\pi$ are integer multiples of π). So $x = k\pi$ is forced and is the entire candidate set.
3. At $x = k\pi$: $\cos x = (-1)^k$, $\cos 2x = 1$, $\cos 3x = (-1)^k$, so the product = $(-1)^k \cdot 1 \cdot (-1)^k = 1$ — every $x = k\pi$ works.
4. In $[0, 2\pi]$ that gives $x = 0, \pi, 2\pi$: $\boxed{3}$ solutions.

Telescoping the modulus

1. For the product to be 1 we need $|\cos x| = |\cos 2x| = |\cos 3x| = 1$ (any factor with $|\cdot| < 1$ would shrink the product below 1 in absolute value).
2. $|\cos x| = 1$ forces $x \in \pi\mathbb{Z}$; $|\cos 2x| = 1$ forces $x \in \frac{\pi}{2}\mathbb{Z}$; $|\cos 3x| = 1$ forces $x \in \frac{\pi}{3}\mathbb{Z}$.
3. The common solutions are $x \in \pi\mathbb{Z}$ (the intersection, since lcm of the moduli forces multiples of π); on these the signs multiply to +1 as shown.
4. Intersecting $\pi\mathbb{Z}$ with $[0, 2\pi]$ yields $\{0, \pi, 2\pi\}$, hence $\boxed{3}$.

INSIGHT

The same 'pin every factor to ± 1 ' logic handles $\prod_{j=1}^n \cos jx = 1$: the constraint $|\cos x| = 1$ (the tightest) forces $x \in \pi\mathbb{Z}$, and the sign product is $\prod (-1)^{jk} = (-1)^{k \cdot n(n+1)/2}$, which equals +1 for all k exactly when $\frac{n(n+1)}{2}$ is even. For $n = 3$ that exponent is $6k$, always even – so every $x = k\pi$ qualifies.

CHAPTER IX

Triangle Geometry Through Trig



Sine rule, cosine rule, area, R , r and s through angles.

№87 Thirteen-Fourteen-Fifteen Radii

●●●●● L4

EVALUATE

For a triangle with sides 13, 14, 15, find R and r .

Heron inradius circumradius

ANSWER

$$R = \frac{65}{8}, \quad r = 4$$

THE TRAP

The trap is using the right-triangle formula $R = c/2$. The triangle 13, 14, 15 is not right.

Method I: The chapter instrument

1. Heron's formula gives $s = 21$ and $\Delta = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$; then $r = \Delta/s$ and $R = abc/(4\Delta)$.
2. Substituting the given restrictions and simplifying gives $R = \frac{65}{8}$, $r = 4$.

Method II: Independent check

1. Compute the area trigonometrically from any angle using the cosine law, then use $\Delta = rs = abc/(4R)$.
2. This route reaches the same value and rules out the trap, so the answer is $R = \frac{65}{8}$, $r = 4$.

INSIGHT

The area is the bridge between side data and radius data.

№88 Included Sixty Degrees

●●●●● L3

EVALUATE

In $\triangle ABC$, $A = 60^\circ$, $b = 7$, $c = 8$. Find a and Δ .

law-of-cosines area included-angle

ANSWER

$$a = \sqrt{57}, \quad \Delta = 14\sqrt{3}$$

THE TRAP

The trap is $a = 15$, adding sides as if the included angle were 180° .

Method I: The chapter instrument

1. Use $a^2 = b^2 + c^2 - 2bc \cos A$ and $\Delta = \frac{1}{2}bc \sin A$.
2. Substituting the given restrictions and simplifying gives $a = \sqrt{57}$, $\Delta = 14\sqrt{3}$.

Method II: Independent check

1. Drop the altitude from C to AB and resolve side c into components using $\sin 60^\circ, \cos 60^\circ$.
2. This route reaches the same value and rules out the trap, so the answer is $a = \sqrt{57}$, $\Delta = 14\sqrt{3}$.

INSIGHT

The included angle controls both the opposite side and the area.

№89 Side From Circumradius

●●●●● L3

EVALUATE

If $R = 5$ and $A = 30^\circ$ in a triangle, find side a .

law-of-sines

circumradius

side

ANSWER

5

THE TRAP

The trap is 10, using the diameter $2R$ but forgetting the sine factor.

Method I: The chapter instrument

1. By the sine rule, $a = 2R \sin A = 10 \cdot 1/2$.
2. Substituting the given restrictions and simplifying gives 5.

Method II: Independent check

1. In the circumcircle, side a is a chord subtending central angle $2A$, hence $a = 2R \sin A$.
2. This route reaches the same value and rules out the trap, so the answer is 5.

INSIGHT

Every side of a triangle is a chord of its circumcircle.

№90 Cosine Sum and Radii

●●●●● L4

PROVE THAT

$$\cos A + \cos B + \cos C = 1 + \frac{r}{R}$$

triangle

radii

cosine

CONCLUSION

$$1 + \frac{r}{R}$$

THE TRAP

The trap is 1, which is true only in a limiting degenerate sense, not for ordinary triangles.

Method I: The chapter instrument

1. Use $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$ and half-angle formulas to reach $1 + 4 \sin(A/2) \sin(B/2) \sin(C/2)$.
2. Substituting the given restrictions and simplifying gives $1 + \frac{r}{R}$.

Method II: Independent check

1. Use $r = 4R \sin(A/2) \sin(B/2) \sin(C/2)$.
2. This route reaches the same value and rules out the trap, so the answer is $1 + \frac{r}{R}$.

INSIGHT

№91 Half-Sine Square Sum

●●●●● L4

EVALUATE

$$\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} \text{ in terms of } r, R.$$

half-angle radii triangle

ANSWER

$$1 - \frac{r}{2R}$$

THE TRAP

The trap is $3/2$, averaging the half-angle squares without using the triangle relation.

Method I: The chapter instrument

1. Use $\sin^2(A/2) = (1 - \cos A)/2$ and sum cyclically.
2. Substituting the given restrictions and simplifying gives $1 - \frac{r}{2R}$.

Method II: Independent check

1. Insert $\cos A + \cos B + \cos C = 1 + r/R$.
2. This route reaches the same value and rules out the trap, so the answer is $1 - \frac{r}{2R}$.

INSIGHT

Once cosine sums are known, half-angle square sums become bookkeeping.

№92 Right Triangle Inradius

●●●●● L3

PROVE THAT

$$\text{In a right triangle with legs } a, b \text{ and hypotenuse } c, \quad r = \frac{a + b - c}{2}.$$

right-triangle inradius area

CONCLUSION

$$\frac{a + b - c}{2}$$

THE TRAP

The trap is $c/2$, confusing inradius with circumradius in a right triangle.

Method I: The chapter instrument

1. Use $\Delta = ab/2$ and $s = (a + b + c)/2$, so $r = \Delta/s = ab/(a + b + c)$; rationalise with $c^2 = a^2 + b^2$.
2. Substituting the given restrictions and simplifying gives $\frac{a+b-c}{2}$.

Method II: Independent check

1. Tangency lengths from the two acute vertices give legs $a = (s - b) + (s - c)$ and $b = (s - a) + (s - c)$, leading to $r = s - c$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{a+b-c}{2}$.

INSIGHT

In a right triangle, the circumradius and inradius have very different formulas.

№93 Half-Angle Tangent Length

●●●●● L4

PROVE THAT

$$\tan \frac{A}{2} = \frac{r}{s-a}$$

half-angle

inradius

tangent-lengths

CONCLUSION

$$\frac{r}{s-a}$$

THE TRAP

The trap is r/s , forgetting that the tangent point cuts off the adjacent side segment $s-a$.

Method I: The chapter instrument

1. In the right triangle formed by the incenter, vertex A , and the touchpoint on AB , the opposite side is r and the adjacent tangent length is $s-a$.
2. Substituting the given restrictions and simplifying gives $\frac{r}{s-a}$.

Method II: Independent check

1. Use $\tan(A/2) = \sin A / (1 + \cos A)$ with $\sin A = 2\Delta/(bc)$ and $1 + \cos A = ((b+c)^2 - a^2)/(2bc)$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{r}{s-a}$.

INSIGHT

The incenter turns half-angle trig into side-segment ratios.

№94 Radii Determine Cosine Sum

●●●●● L3

EVALUATE

If $R = 10$ and $r = 4$, evaluate $\cos A + \cos B + \cos C$.

radii

cosine-sum

triangle

ANSWER

$$\frac{7}{5}$$

THE TRAP

The trap is 1, forgetting the inradius correction.

Method I: The chapter instrument

1. Use $\cos A + \cos B + \cos C = 1 + r/R$.
2. Substituting the given restrictions and simplifying gives $\frac{7}{5}$.

Method II: Independent check

1. Use the equivalent half-angle product formula $1 + 4 \sin(A/2) \sin(B/2) \sin(C/2)$ and $r = 4R$ times that product.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{7}{5}$.

INSIGHT

Radius ratios can encode surprisingly direct angle sums.

№95 A Double-Angle Triangle

●●●●● L4

SOLVE

In $\triangle ABC$, $A = 2B$ and $a : b = \sqrt{3} : 1$. Find (A, B, C) .

law-of-sines

double-angle

triangle

SOLUTIONS

$$(A, B, C) = \left(\frac{\pi}{3}, \frac{\pi}{6}, \frac{\pi}{2}\right)$$

THE TRAP

The trap is $A = 120^\circ, B = 60^\circ$, which leaves no positive angle for C .

Method I: The chapter instrument

1. By the sine rule, $a/b = \sin 2B / \sin B = 2 \cos B = \sqrt{3}$, so $B = \pi/6$.
2. Substituting the given restrictions and simplifying gives $(A, B, C) = (\frac{\pi}{3}, \frac{\pi}{6}, \frac{\pi}{2})$.

Method II: Independent check

1. Then $A = 2B = \pi/3$ and $C = \pi - A - B = \pi/2$.
2. This route reaches the same value and rules out the trap, so the answer is $(A, B, C) = (\frac{\pi}{3}, \frac{\pi}{6}, \frac{\pi}{2})$.

INSIGHT

Side ratios plus angle multiples usually collapse through the sine rule.

№96 Three Sine-Rule Copies

●●●●● L3

EVALUATE

$$\frac{a}{\sin A} + \frac{b}{\sin B} + \frac{c}{\sin C}$$

sine-rule

circumradius

sum

ANSWER

$$6R$$

THE TRAP

The trap is $2R$, remembering one sine-rule ratio but forgetting there are three identical terms.

Method I: The chapter instrument

1. By the sine rule, each term equals $2R$.
2. Substituting the given restrictions and simplifying gives $6R$.

Method II: Independent check

1. View each side as a chord: $a = 2R \sin A, b = 2R \sin B, c = 2R \sin C$.
2. This route reaches the same value and rules out the trap, so the answer is $6R$.

INSIGHT

The sine rule often converts a whole sum into repeated constants.

CHAPTER X

Grand Hybrids

Cyclotomic sums, Chebyshev, substitution and inequalities fused.



№97 Chebyshev Root Filter

●●●●● L5

EVALUATE

$$\sum_{k=0}^{n-1} T_m \left(\cos \frac{2\pi k}{n} \right)$$

Chebyshev

roots-of-unity

filter

ANSWER

$$\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

THE TRAP

The trap is 0 always, forgetting that $T_m(\cos \theta) = \cos m\theta$ can resonate with the n -point grid.

Method I: The chapter instrument

1. Use $T_m(\cos \theta) = \cos m\theta$, then apply the finite cosine frequency filter.
2. Substituting the given restrictions and simplifying gives $\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$

Method II: Independent check

1. Write $T_m((z + z^{-1})/2) = (z^m + z^{-m})/2$ on the n th roots and sum powers.
2. This route reaches the same value and rules out the trap, so the answer is $\begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$

INSIGHT

This is exactly Chebyshev fused with roots-of-unity orthogonality.

№98 Dyadic Product at a Sharp Angle

●●●●● L4

EVALUATE

$$\prod_{k=0}^{m-1} \cos \frac{2^k \pi}{2^{m+1}}$$

dyadic

product

exact-value

ANSWER

$$\frac{1}{2^m \sin \frac{\pi}{2^{m+1}}}$$

THE TRAP

The trap is 0, because the final doubled angle is near $\pi/2$. It reaches $\pi/2$ only after the product identity has put sine in the numerator.

Method I: The chapter instrument

1. Apply $\prod_{k=0}^{m-1} \cos(2^k x) = \sin(2^m x) / (2^m \sin x)$ with $x = \pi/2^{m+1}$.
2. Substituting the given restrictions and simplifying gives $\frac{1}{2^m \sin \frac{\pi}{2^{m+1}}}$.

Method II: Independent check

1. Induct using $2 \sin y \cos y = \sin 2y$ until the numerator becomes $\sin(\pi/2)$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{1}{2^m \sin \frac{\pi}{2^{m+1}}}$.

INSIGHT

Endpoint angles are where telescoping products become exact constants.

№99 Rational Range by Half-Angle

••••• L4

FIND THE RANGE

$$\frac{3+4t}{1+t^2} \quad (t \in \mathbb{R})$$

substitution

range

discriminant

RANGE

$$[-1, 4]$$

THE TRAP

The trap is $[-5, 5]$, treating the numerator as an unrestricted dot product without the denominator.

Method I: The chapter instrument

1. Set the value equal to y and require $yt^2 - 4t + (y-3) = 0$ to have real t .
2. Substituting the given restrictions and simplifying gives $[-1, 4]$.

Method II: Independent check

1. Put $t = \tan(x/2)$; the expression becomes $\frac{3}{2}(1 + \cos x) + 2 \sin x$ and has the same shifted sinusoidal range.
2. This route reaches the same value and rules out the trap, so the answer is $[-1, 4]$.

INSIGHT

Half-angle substitutions and discriminants are two views of the same range test.

№100 Cyclotomic Chord With Parameter

••••• L5

EVALUATE

$$\prod_{k=1}^{n-1} \left(1 - 2a \cos \frac{2\pi k}{n} + a^2 \right) \quad (a \neq 1)$$

roots-of-unity

product

parameter

ANSWER

$$\left(\frac{a^n - 1}{a - 1} \right)^2$$

THE TRAP

The trap is $a^{2n} - 1$, multiplying roots without removing the missing $k = 0$ factor.

Method I: The chapter instrument

1. Write each factor as $(a - \omega^k)(a - \omega^{-k})$ and multiply over $k = 1, \dots, n-1$.
2. Substituting the given restrictions and simplifying gives $\left(\frac{a^n - 1}{a - 1} \right)^2$.

Method II: Independent check

1. Use $\prod_{k=1}^{n-1} (a - \omega^k) = (a^n - 1)/(a - 1)$, then take the reciprocal-index copy.
2. This route reaches the same value and rules out the trap, so the answer is $\left(\frac{a^n-1}{a-1}\right)^2$.

INSIGHT

A parameter turns a chord product into a cyclotomic polynomial evaluation.

№101 Triangle Product Capstone

●●●●● L4

FIND THE MAXIMUM

$$A, B, C > 0, A + B + C = \pi : \sin A \sin B \sin C$$

triangle

inequality

product

MAXIMUM

$$\frac{3\sqrt{3}}{8}$$

THE TRAP

The trap is 1, maximising the three factors separately.

Method I: The chapter instrument

1. Use the conditional-angle constraint with Jensen on $\log \sin x$.
2. Substituting the given restrictions and simplifying gives $\frac{3\sqrt{3}}{8}$.

Method II: Independent check

1. Use the substitution $x = \tan(A/2)$, $y = \tan(B/2)$, $z = \tan(C/2)$ with $xy + yz + zx = 1$ and smooth to $x = y = z = 1/\sqrt{3}$.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{3\sqrt{3}}{8}$.

INSIGHT

This fuses the half-angle instrument with the inequality instrument.

№102 Derivative of the Root Filter

●●●●● L5

EVALUATE

$$\sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n}$$

roots-of-unity

derivative

cosecant

ANSWER

$$\frac{n^2 - 1}{3}$$

THE TRAP

The trap is $n - 1$, as if every sine were 1.

Method I: The chapter instrument

1. Compare the coefficient of x in the expansion $n \cot nx - \cot x = \sum_{k=1}^{n-1} (\cot(x + k\pi/n) - \cot(k\pi/n))$ near 0.
2. Substituting the given restrictions and simplifying gives $\frac{n^2-1}{3}$.

Method II: Independent check

1. Differentiate the logarithmic derivative of $(z^n - 1)/(z - 1)$ and translate $|1 - \omega^k|^2$ into sine squares.
2. This route reaches the same value and rules out the trap, so the answer is $\frac{n^2-1}{3}$.

INSIGHT

This is a root-of-unity identity with a calculus edge.

№103 Cosine Five Meets Cosine Three

●●●●● L4

SOLVE

$$\cos 5x = \cos 3x \quad (0 \leq x < 2\pi)$$

equation

product-to-sum

multiple-angle

SOLUTIONS

$$\left\{ \frac{k\pi}{4} : 0 \leq k \leq 7 \right\}$$

THE TRAP

The trap is four roots, from solving only $5x = 3x$ and ignoring $5x = -3x$ modulo 2π .

Method I: The chapter instrument

1. Use $\cos 5x - \cos 3x = -2 \sin 4x \sin x$ and merge the zeros.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{k\pi}{4} : 0 \leq k \leq 7 \right\}$.

Method II: Independent check

1. Use $\cos u = \cos v$: $u = \pm v + 2k\pi$, then restrict to the interval.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{k\pi}{4} : 0 \leq k \leq 7 \right\}$.

INSIGHT

The product-to-sum form automatically handles both cosine branches.

№104 Triangle Side Projection

●●●●● L4

PROVE THAT

$$\text{In } \triangle ABC, \quad a \cos B + b \cos A = c.$$

geometry

sine-rule

conditional

CONCLUSION

$$c$$

THE TRAP

The trap is $a + b$, projecting both sides in the wrong direction without the angle weights.

Method I: The chapter instrument

1. Use $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$; then the left side is $2R(\sin A \cos B + \sin B \cos A)$.
2. Substituting the given restrictions and simplifying gives c .

Method II: Independent check

1. Since $A + B = \pi - C$, the sine addition formula gives $\sin(A + B) = \sin C$.
2. This route reaches the same value and rules out the trap, so the answer is c .

INSIGHT

This is triangle geometry reduced to one conditional-angle identity.

№105 Chebyshev Bound

●●●●● L5

PROVE THAT

$$|x| \leq 1 \implies |T_n(x)| \leq 1.$$

Chebyshev inequality range

CONCLUSION

$$|T_n(x)| \leq 1$$

THE TRAP

The trap is using the leading term $2^{n-1}x^n$ and predicting huge values near 1. The lower terms are essential on $[-1, 1]$.

Method I: The chapter instrument

1. Set $x = \cos \theta$; then $T_n(x) = \cos n\theta$, whose absolute value is at most 1.
2. Substituting the given restrictions and simplifying gives $|T_n(x)| \leq 1$.

Method II: Independent check

1. Use the extremal interpretation of Chebyshev polynomials on $[-1, 1]$: their graph oscillates between -1 and 1 at the alternation points.
2. This route reaches the same value and rules out the trap, so the answer is $|T_n(x)| \leq 1$.

INSIGHT

The substitution $x = \cos \theta$ is the whole reason Chebyshev polynomials are useful.

№106 Dyadic Product Equation

●●●●● L5

SOLVE

$$\prod_{k=0}^{m-1} 2 \cos(2^k x) = 1 \quad \left(0 < x \leq \frac{\pi}{2^m}\right)$$

dyadic equation telescope

SOLUTIONS

$$\left\{ \frac{\pi}{2^m + 1} \right\}$$

THE TRAP

The trap is $x = 0$, suggested by a limiting value; the interval excludes it and the product is not exactly 1 there as a point.

Method I: The chapter instrument

1. Use $\prod_{k=0}^{m-1} 2 \cos(2^k x) = \sin(2^m x) / \sin x$, so the equation becomes $\sin(2^m x) = \sin x$.
2. Substituting the given restrictions and simplifying gives $\left\{ \frac{\pi}{2^m + 1} \right\}$.

Method II: Independent check

1. On the stated interval, $2^m x \in (0, \pi]$, so the nonzero solution branch is $2^m x = \pi - x$.
2. This route reaches the same value and rules out the trap, so the answer is $\left\{ \frac{\pi}{2^m + 1} \right\}$.

INSIGHT

This is a telescope, an equation, and a domain restriction in one problem.